Finite-dimensional descent and moment-kernel methods for published questions

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Complete arguments for review; historical priority not certified

Abstract

We present six mathematical arguments addressing the original statements of published questions in symbolic integration, total positivity, special functions, and probability. A finite-dimensional descent through algebraic function-field layers shows that elementary-stable functions are D-finite. Combined with a finite adjoint-moment criterion, it gives a proposed complete characterization for Chen–Kauers Problem 8. A second general result characterizes coefficientwise Hankel total positivity of exponential Bell moment polynomials by the Stieltjes property of their shifted cumulants. A proof using positive semidefinite formal kernels and Schur positivity applies to Sokal's functional-digraph Conjecture 6.9(b); an explicit Lambert-function measure also proves part (d). A covariance identity proves Yang's separately stated DeTemple-quotient monotonicity conjecture. A matrix marginal theorem supplies a gamma-mixture representation for an orthogonal-polynomial transform. Applied to the DeTemple spectral weight, it gives the coefficients and strict complete monotonicity at every order in Yang–Chu–Zhang Problem 1. A boundary profile, with an elementary explicit interior certificate, refutes the Ferreira–Simon Wright log-concavity conjecture. Finally, a negative third cumulant gives a counterexample to the universal negative-power question of Yano, Yano and Yor. These are internally checked research arguments, not a claim to have resolved ten difficult open conjectures. Literature searches were bounded, and historical priority remains subject to further review. The revision also proves that the space of elementarily integrable polynomial weights is finite-dimensional or finite-codimensional, according as the elementary function is not or is D-finite. Chen's remaining moment-set conjecture is not claimed solved. The document includes a short classical proof of Sokal's Conjecture 6.11(a) and three exact obstructions to broader positivity-preservation arguments. These additions are not counted as new difficult-conjecture resolutions.

1 Scope, original targets, and status

The research objective was to obtain ten new unconditional resolutions of pre-existing difficult conjectures, keeping each original statement unchanged. That objective has *not* been reached. This document preserves six proof candidates in a form that can be checked independently. It does not certify that all six remained open at the date of writing, that their value exceeds the earlier corpus, or that each uses a historically new principle. Those assertions require evidence beyond the proofs themselves.

The term *total positivity* follows the sources: all minors are nonnegative, rather than strictly positive. A universal conjecture is refuted by a counterexample in its original domain; such a refutation need not classify every parameter in that domain. We use this distinction only where the original source actually asks a universal question.

Original target	Conclusion established by the argument below
Chen–Kauers, Problem 8	Elementary stability implies D-finiteness; an adjoint-moment criterion gives a necessary and sufficient finite-test classification for the original class.
Sokal, Conjecture 6.9(b),(d)	Coefficientwise Hankel total positivity of the functional-digraph polynomials, and a positive moment measure for normalized connected counts. The two clauses are treated as one target.
Yang, Conjecture 1(i)	Strict increase of the original quotient on its whole domain, with both sharp endpoint limits.
Yang–Chu–Zhang, Problem 1	All-order coefficient determination, strict complete monotonicity in the original variable x, the required nonzero asymptotic constant, and uniqueness up to positive scaling.
Ferreira–Simon, Section 6.4	Failure of the proposed sufficient threshold for Wright-density log-concavity, including a strictly interior rational parameter pair.
Yano–Yano–Yor, Remark 2.9(iii)	A negative power of the original half-α-Cauchy variable that is not infinitely divisible and therefore not self-decomposable.

The first proposed methodological contribution is a uniform bound on elementary primitives modulo a lower function field, followed by finite-dimensional differential-module descent. It handles arbitrary finite algebraic adjunctions, including ramification. A second contribution is the passage from a shifted-cumulant moment condition to *coefficientwise* positivity of every Bell Hankel minor. Ordinary positive definiteness at each positive parameter is not, by itself, a coefficientwise argument. The proof resolves this issue by introducing auxiliary polynomial weights and extracting Schur coefficients. The new all-order argument uses a Schur-complement matrix marginal to obtain a gamma-mixture law, and then a divided-difference identity to prove positivity of the exact Laplace kernel. The three other special-function and probability arguments use classical covariance, boundary-limit, and cumulant tools in specific applications. They should not be described as three new foundational principles.

The source corpus was inspected at commit `ecd87de3348dc` of Verifiable Mathematical Reasoning Corpus. It served as a research starting point, not as an authority certifying its own claims. None of the arguments below assumes that an earlier corpus claim is a proved theorem. A previously prepared Le Roy counterexample is not counted again here. Partial graph results, finite searches, and unproved extensions are also excluded from the table.

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2 Elementary stability and a finite adjoint criterion

Status and scope. This section presents a proposed complete proof for review of a structural classification answering Chen–Kauers Problem 8. It is not a claim of independently certified historical priority. The problem asks for the class of elementary functions whose every repeated indefinite integral is elementary [19, Problem 8]. We prove that such functions must be D-finite, and then give a finite elementary-integrability criterion once an annihilator is known. The proof allows arbitrary finite algebraic adjunctions interspersed with logarithmic and exponential adjunctions.

Throughout, C is an algebraically closed field of characteristic zero, $B = C(x)$, and $Dx = 1$. All differential fields have constants C. For ordinary complex elementary functions one takes $C = \mathbb{C}$ and works with compatible germs in a common differential extension; choices of integration constants do not affect stability. A function is *elementarily stable* if it has an elementary n-th primitive for every $n \geq 1$.

Chen established the moment formulation and several special families of stable functions [20]. Sahu, Seiss and Srinivasan gave a generalized Liouville theorem for an elementary primitive of any fixed depth [21]. The complete reduction of Chen et al. treats single-step integration in transcendental Liouvillian towers [22]. The argument here uses the classical Liouville theorem [23] and Rosenlicht's local function-field estimates [26], together with a uniform relative bound: elementary primitives of integrands in a fixed finite-dimensional B-space lie in one finite-dimensional B-space modulo a lower field. A finite-dimensional differential-module argument then removes one transcendental layer. The same bound gives a finite/cofinite dichotomy for integrable polynomial weights. Chen's separate moment-set conjecture remains unresolved in its D-finite case.

2.1 Moments and a finite primitive-space lemma

Lemma 2.1 (Moment criterion)**.** *An elementary function f is stable if and only if $x^n f$ has an elementary primitive for every $n \geq 0$. The stable functions form a $C[x][D]$-module.*

Proof. Repeated integration by parts expresses a primitive of $x^n f$ in terms of the first $n+1$ primitives of f. Conversely, if $Dg_k = x^k f$ for $0 \leq k < n$, then

$$\frac{1}{(n-1)!} \sum_{k=0}^{n-1} (-1)^k \binom{n-1}{k} x^{n-1-k} g_k$$

is an n-th primitive of f. Addition and multiplication by $C[x]$ preserve the moment condition. Differentiation preserves it because $x^n Df = D(x^n f) - nx^{n-1} f$, with the second term absent for $n = 0$. □

Lemma 2.2 (Finite primitive space for independent weights)**.** *Let M be a differential B-module. Suppose that $f, g_0, \ldots, g_d \in M$, that the g_i span a B-space of dimension at most d, and that*

$$Dg_i = p_i f \qquad (0 \leq i \leq d),$$

where $p_0, \ldots, p_d \in B$ are linearly independent over C. Then f belongs to $V = \operatorname{span}_B(g_0, \ldots, g_d)$, $DV \subseteq V$, and f is annihilated by a nonzero operator in $B[D]$.

Proof. Suppose $f \notin V$. Choose a nonzero relation $\sum_{i \in J} a_i g_i = 0$ with J of minimal cardinality, and normalize one coefficient to 1. Differentiating gives

$$\sum_{i \in J} a_i' g_i + \left(\sum_{i \in J} a_i p_i \right) f = 0.$$

Since $f \notin V$, both terms vanish separately. The first relation has smaller support unless every $a_i' = 0$, so minimality forces all $a_i \in C$. The second relation then contradicts the C-linear independence of the p_i. Hence $f \in V$. The equations $Dg_i = p_i f$ imply $DV \subseteq V$, and finite dimension gives a linear dependence among successive derivatives of f. □

2.2 One algebraic function-field layer

We use strong induction on the number of transcendental generators in a finite Liouvillian tower over B. Arbitrary finite algebraic adjunctions are allowed and are included in the induction hypothesis. At a positive level write

$$E/F(t)/F/B, \tag{2.1}$$

where $E/F(t)$ is finite algebraic, F has one fewer transcendental generator, t is transcendental, and either

$$Dt = \beta \in F \quad \text{or} \quad Dt = \alpha t, \quad \alpha \in F.$$

We call these the primitive and exponential cases. Replacing F by its relative algebraic closure in E is a finite algebraic enlargement and does not increase the induction level. We do so, and view E as the function field of a smooth projective geometrically integral curve X/F, finite over the t-line.

Lemma 2.3 (Properness after algebraic enlargement)**.** *In the primitive case, β is not a derivative in any algebraic extension of F. In the exponential case, no nonzero rational multiple of α is a logarithmic derivative of an element algebraic over F.*

Proof. If $Du = \beta$ in a finite algebraic extension F_1/F, then normalized trace gives a primitive in F. Subtracting it from t gives a new constant, contrary to the hypotheses. If $Du/u = q\alpha$ with $q = a/b \in \mathbb{Q} \setminus \{0\}$ and u algebraic over F, then $D(u^b/t^a) = 0$. The field containing this quotient is algebraic over $F(t)$, so its constants are still C. This would make a nonzero power of t algebraic over F, a contradiction. Here we use the standard fact that constants in an algebraic differential extension are algebraic over the original constant field; C is algebraically closed. □

An irreducible $p \in F[t]$ of positive degree is called normal when $p \nmid Dp$. Every finite irreducible is normal in the primitive case. Indeed, for monic p of degree d, divisibility would imply $Dp = 0$; the coefficient of t^{d-1} would give $Db = -d\beta$ for the next coefficient b, contradicting Lemma 2.3. In the exponential case the same argument shows that every irreducible other than t is normal: a monic special polynomial of degree d with nonzero constant term b_0 would satisfy $Db_0 = d\alpha b_0$.

The local valuation arguments are classical. Rosenlicht proved continuity of the derivation at function-field places and corresponding ramified-place estimates for derivatives and logarithmic derivatives in primitive and exponential extensions [26, Lemmas 1–2]. We record the bounds needed here, together with the fixed finite coefficient extension used in the subsequent uniform argument.

Lemma 2.4 (Local bounds and Puiseux coordinates)**.** *Let P lie over a finite normal divisor p, with ramification index e, and normalize its valuation v_P to have value group $\mathbb{Z}$. Then*

$$v_P(Dh) = v_P(h) - e \qquad \text{if } v_P(h) < 0, \tag{2.2}$$

$$v_P(Dh) \geq 1 - e \qquad \text{if } h \text{ is a local unit}, \tag{2.3}$$

$$v_P(Du/u) \geq -e \qquad \text{if } u \in E^*. \tag{2.4}$$

At a selected place over infinity, a fixed finite algebraic extension F'/F gives an embedding of the base-changed function field into $F'((t^{-1/e}))$ that commutes with D. Analogous local expansions hold at all the other places under consideration.

Proof. For the finite-place assertions, extend algebraic coefficients and choose a root c of p. Since characteristic is zero, $p_t(c) \neq 0$, and differentiation of $p(c) = 0$ gives

$$(Dp)(c) = p_t(c)((Dt)|_{t=c} - Dc).$$

Thus $\gamma = (Dt)|_{t=c} - Dc$ is nonzero. In a Puiseux coordinate z with $t - c = z^e$,

$$Dz = \frac{\gamma}{e} z^{1-e} + \text{terms of degree at least } 1.$$

For $h = \sum_{k \geq r} h_k z^k$, $h_r \neq 0$, and $r < 0$, the leading term of Dh is $(r\gamma/e)h_r z^{r-e}$. Derivatives of coefficients have exponent at least r, and derivatives of higher powers have exponent at least $r + 1 - e$. This proves (2.2). For a unit, differentiating its constant coefficient is regular and every positive power contributes exponent at least $1 - e$, proving (2.3). Factoring u into a power of z times a unit gives (2.4).

For the finite coefficient extension, first extend F by the finite residue field of a selected point over infinity. The corresponding point after smooth base change is rational. Its completed local ring is $F_1[[z]]$, with the given F_1 as coefficient field. Write $t^{-1} = z^e u(z)$, where $u(0) \neq 0$. Adjoin an e-th root of $u(0)$, a finite extension of F_1. Hensel lifting, or the formal binomial expansion in characteristic zero, supplies an e-th root $U(z)$ of $u(z)$ over the resulting finite field F'. Then $w = zU(z)$ is a uniformizer with $w^e = t^{-1}$, and the completed field is $F'((w))$. The coefficientwise formal derivation having the specified value on t agrees on the algebraic

function field with the unique extension of D. The finite-place constructions are the same after replacing t^{-1} by a local base coordinate. For local bounds one may work in an algebraic closure of F; only one selected place needs the fixed finite coefficient extension for the global argument. □

For an expansion at infinity, write $\deg_P h = d$ when its leading term is $h_d t^d$, with $d \in e^{-1}\mathbb{Z}$. A rational point over infinity also supplies a useful basis construction. For any fixed divisor H, the Riemann–Roch space $\mathcal{L}(H)$ is finite-dimensional over F'. Its valuation filtration at that point has graded pieces of dimension at most one. It therefore has a basis $w_1, \dots, w_s$ with distinct leading degrees $d_1 > \cdots > d_s$, each leading coefficient normalized to 1. In the primitive case $\deg_P Dw_j < d_j$; in the exponential case the degree-d_j coefficient of Dw_j is $d_j\alpha$. If $1 \in \mathcal{L}(H)$, the basis can include 1 as its unique degree-zero element: subtract the leading constant from every other element initially having degree zero.

2.3 Uniform first-order preimages

Lemma 2.5 (Uniform preimage space)**.** *Let K/B be a finite Liouvillian tower with no new constants, allowing arbitrary finite algebraic adjunctions. For $a \in K$ and a finite-dimensional B-space $V \subseteq K$, the set*

$$\{g \in K : Dg + ag \in V\}$$

is contained in a finite-dimensional B-subspace of K.

Proof. Use strong induction on the transcendental length. At length zero K/B is finite algebraic, and the assertion is immediate. At positive length use (2.1), with $K = E$.

We first bound the pole divisor of every solution g. At a finite normal place P, let M bound the pole orders of generators of V, and suppose $q = v_P(g) < 0$. If $v_P(a) < -e$, multiplication by a dominates differentiation, so $q + v_P(a) \geq -M$. If $v_P(a) > -e$, differentiation dominates, so $q - e \geq -M$. If $v_P(a) = -e$, the local leading coefficient is

$$h_q(q\gamma/e + a_{-e}).$$

It can vanish for at most one integer q. This gives a finite bound in every case. Outside the finite set consisting of poles of a, poles of generators of V, and places over branch points, the bound forces g to be regular.

Next consider every place above infinity. Set $A = \deg_P a$, and let N bound the degrees of generators of V. When $A > 0$, multiplication dominates because differentiation does not increase Puiseux degree. Consequently $d = \deg_P g \leq N - A$.

In the exponential case with $A \leq 0$, a solution with $d > N$ must have leading coefficient satisfying

$$\frac{Dh_d}{h_d} + d\alpha + a_0 = 0.$$

There is at most one rational d for which this is possible with h_d algebraic over F: two choices contradict Lemma 2.3 after dividing their leading coefficients. Thus d is bounded. At each place over zero the same argument uses the lowest Puiseux exponent. A pole of a gives a direct lower bound on that exponent; otherwise there is at most one exceptional exponent below those occurring in V.

In the primitive case with $A \leq 0$, suppose $d > N + 1$. Divide the equation by g and write, purely formally,

$$g = h_d t^d \exp(\Lambda), \qquad \Lambda = \log(1 + \text{strictly lower Puiseux powers}).$$

The right side of $Dg/g + a = v/g$ has degree below -1. If Λ_1 is the coefficient of t^{-1} in Λ, comparison at t^{-1} gives

$$d\beta + D\Lambda_1 + a_{-1} = 0. \tag{2.5}$$

Indeed, $\beta\partial_t\Lambda$ has degree strictly below -1, even when fractional exponents occur. Two distinct rational values of d satisfying (2.5) would give a primitive of β algebraic over F, excluded by Lemma 2.3. There is at most one exceptional degree. This proves a bound at every infinity place, not merely at the place chosen later for coefficient extraction.

The local bounds yield a fixed divisor H containing every pole divisor of g. Make one infinity place rational over a fixed finite extension F'/F and choose the normalized echelon basis of $\mathcal{L}(H)$ described above. Write $g = \sum h_j w_j$ with $h_j \in F'$. If a has positive degree at that place, the highest coefficient of ag determines h_1 algebraically from a coefficient of $v \in V$, so h_1 lies in a fixed finite B-space. Otherwise the highest coefficient equation has one of the forms

$$Dh_1 + a_0 h_1 \in V_1, \qquad Dh_1 + (a_0 + d_1\alpha)h_1 \in V_1,$$

with $V_1 \subset F'$ a fixed finite B-space. The induction hypothesis applies because the finite algebraic enlargement F' has smaller transcendental length.

Subtract $h_1 w_1$ and descend through the finite basis. Each new right side remains in a fixed finite B-space: the derivative of a finite B-space is contained in a finite B-space, since $DB \subseteq B$. Induction bounds every h_j, and hence g, in a finite B-space inside EF'. Intersecting that space with E proves the assertion. The bounds used here are existential; their effectivity is not needed. □

2.4 Moving logarithms and descent modulo the lower field

Lemma 2.6 (Relative elementary primitive space)**.** *In (2.1), let $V \subset E$ be a finite-dimensional B-space. There exist a fixed finite algebraic extension F'/F, an elementary extension A of EF', and a fixed finite-dimensional B-space $W \subset A$ such that every elementarily integrable $v \in V$ admits*

$$v = Dw + q, \qquad w \in W, \quad q \in F'.$$

The fields and the space are independent of v.

Proof. Liouville's theorem gives

$$v = Dg + \sum_j c_j \frac{Du_j}{u_j}, \qquad g \in E, \quad u_j \in E^*, \quad c_j \in C. \tag{2.6}$$

The version with all terms in E follows from the usual algebraic version by trace for the rational part and norm for logarithmic derivatives.

Let S contain the poles of generators of V, all places over branch points, all infinity places, and all places above zero in the exponential case. At $P \notin S$, the derivation is regular and transverse. A pole of g would give a pole of Dg of order at least two, whereas logarithmic derivatives have pole order at most one and v is regular. Hence g is regular there. The simple-pole coefficient then forces

$$\sum_j c_j \operatorname{ord}_P(u_j) = 0.$$

Thus the formal divisor combination $\sum_j c_j \operatorname{div}(u_j)$ is supported on the fixed finite set S.

These combinations form a C-subspace of C^S. Choose a basis of its image and one fixed logarithmic representative for each basis element, writing

$$\ell_i = \sum_k c_{ik} \log u_{ik}, \qquad \delta_i = D\ell_i \in E.$$

If a C-linear logarithmic combination has zero divisor combination, choose a $\mathbb{Q}$-basis of the finite $\mathbb{Q}$-span of its coefficients and clear denominators. Each resulting integer product of the u_j has divisor zero. Since X is projective and F is relatively algebraically closed in E, that product belongs to F^*. Its logarithmic derivative belongs to F. Therefore (2.6) can be written

$$v = Dg + \sum_i \lambda_i \delta_i + q_0, \qquad \lambda_i \in C, \quad q_0 \in F. \tag{2.7}$$

This controls the moving logarithms modulo the lower field, rather than assuming their arguments were fixed initially.

We next bound the pole divisor of the original rational part g. At a finite normal place of ramification e, a pole of g of order m produces a pole of Dg of order $m + e$, whereas all logarithmic derivatives have pole order at most e. The fixed pole bound for v bounds m. At exponential zero and infinity, logarithmic derivatives have respectively nonnegative lowest exponent and nonpositive degree. A nonzero leading exponent of g cannot disappear on differentiation, by Lemma 2.3. This bounds its poles there. At primitive infinity, logarithmic derivatives have degree at most zero. If N is the maximum of zero and the degrees of generators of V, then Dg has degree at most N. A degree $d > N + 1$ would force $d\beta + D\Lambda_1 = 0$ by the t^{-1} coefficient of Dg/g, contradicting Lemma 2.3. These estimates apply at every special place. Together with regularity outside S, they put every g in one fixed Riemann–Roch space.

Make a selected infinity place rational over fixed finite F'/F. Enlarge the bounded-divisor space to include 1, and choose an echelon basis with normalized leading coefficients, distinct degrees, and 1 as its unique degree-zero element. Descend through this basis in (2.7). At every nonzero degree, $q_0 \in F'$ contributes no coefficient. The leading coefficient h_j of g satisfies a first-order equation over F' with coefficient 0 in the primitive case or $d_j\alpha$ in the exponential case, and forcing in a fixed finite B-space. Lemma 2.5 at the smaller transcendental length confines h_j to a finite B-space. At degree zero subtract its unrestricted coefficient h_0 and absorb Dh_0 into q_0. Continue with all negative degrees; the lower-field term still contributes no coefficient there. Consequently

$$g = w_* + h, \qquad h \in F',$$

where w_* ranges in a fixed finite B-space. Adjoin the finitely many logarithms occurring in the fixed ℓ_i and set $w = w_* + \sum_i \lambda_i \ell_i$. Then $v = Dw + q$ for some $q \in F'$, and all such w lie in one fixed finite-dimensional B-space. □

Theorem 2.7 (D-finiteness of elementary-stable functions)**.** *Every elementary-stable function over $C(x)$ is D-finite over $C(x)$.*

Proof. Induct on the transcendental length of an elementary field containing f. At length zero the function is algebraic and hence D-finite. At positive length use (2.1) and apply Lemma 2.6 to $V = Bf$. By Lemma 2.1, every $x^n f$ is elementarily integrable. There are w_n in one fixed finite B-space such that

$$Dw_n = x^n f - q_n, \qquad q_n \in F'.$$

The quotient A/F' is a differential B-module, since F' contains B and is stable under D. Lemma 2.2, applied in this quotient to finitely many of the w_n, gives a nonzero $L \in B[D]$ with $Lf \in F'$. Clear denominators and take $L \in C[x][D]$. This step is necessary because stability is not being asserted to be preserved by arbitrary rational multiplication.

Lemma 2.1 makes Lf elementary-stable. The field F' has smaller transcendental length, with only a finite algebraic enlargement, so induction makes Lf D-finite. A nonzero polynomial-coefficient annihilator M of Lf gives the nonzero annihilator ML of f. □

2.5 A finite criterion for the original classification problem

Let $L = \sum_{j=0}^{r} a_j(x)D^j \neq 0$ annihilate f, with $a_j \in C[x]$. Expand its formal adjoint with coefficients on the left:

$$P = L^* = \sum_{j=0}^{r}(-D)^j a_j(x) = \sum_{i=0}^{r} b_i(x)D^i.$$

Over nonzero b_i, define

$$s = \max_i(\deg b_i - i), \qquad \rho(z) = \sum_{\deg b_i - i = s} \mathrm{lc}(b_i)(z)_i, \tag{2.8}$$

where $(z)_i = z(z-1)\cdots(z-i+1)$. The polynomial ρ is nonzero, because the represented falling factorials have distinct degrees. Choose $N \geq 0$ with $N + s \geq 0$ and $\rho(n) \neq 0$ for every integer $n \geq N$, and put $m = N + s$.

Proposition 2.8 (Finite adjoint-moment criterion)**.** *For an elementary f annihilated by L, stability is equivalent to elementary integrability of*

$$f,\ xf,\ \ldots,\ x^{m-1}f.$$

When $m = 0$, the test list is empty.

Proof. For $n \geq N$, the leading term of $P(x^n)$ is $\rho(n)x^{n+s}$. Triangular elimination by degree gives

$$C[x] = P(C[x]) + \mathrm{span}_C\{1, x, \ldots, x^{m-1}\}. \tag{2.9}$$

Only a vector-space sum is used; $P(C[x])$ is not asserted to be an ideal. The Lagrange identity is

$$qL(f) - fL^*(q) = DJ_L(f, q),$$

with the explicit concomitant

$$J_L(f, q) = \sum_{j=1}^{r}\sum_{k=0}^{j-1}(-1)^k D^k(a_j q)D^{j-1-k}f.$$

Since $Lf = 0$, $fP(q)$ is the derivative of the elementary function $-J_L(f, q)$. Equation (2.9) therefore reduces elementary integrability of every polynomial multiple of f to the displayed finite list. Lemma 2.1 completes the proof. □

Theorem 2.9 (Classification by an annihilator and finite tests)**.** *An elementary function f is stable if and only if it is D-finite and, for one nonzero polynomial-coefficient annihilator, the finite tests of Proposition 2.8 pass.*

Proof. Necessity of D-finiteness is Theorem 2.7; the remaining equivalence is Proposition 2.8. □

Remark 2.10 (One elementary primitive is not enough). Take $f = 2x\exp(x^2)$. It is D-finite and has the elementary primitive $\exp(x^2)$. An annihilator and its adjoint are

$$L = xD - (2x^2 + 1), \qquad L^* = -xD - 2x^2 - 2.$$

Thus $s = 2$, $\rho = -2$, $N = 0$, and the criterion tests both f and xf. The first passes, while the second fails because

$$\int 2x^2 e^{x^2}\,dx = xe^{x^2} - \int e^{x^2}\,dx,$$

and the last integral is not elementary. The finite criterion therefore distinguishes stability from D-finiteness plus one elementary integral.

Remark 2.11 (Effective decision and constant-field assumptions). Assume a proper elementary-field presentation with effective arithmetic, zero tests, the requisite algebraic representations, and an elementary- integrability decision procedure. Also assume that a bound beyond all nonnegative integer zeros of the polynomial ρ can be computed; this is automatic for effectively represented algebraic complex coefficients, and follows from any certified complex-root bound. These are effective input assumptions of symbolic integration; no algorithmic claim is made for arbitrary unrepresented complex constants.

Dovetail two searches. The first tests $f, xf, x^2f, \dots$ and returns "not stable" at the first nonintegrable moment. The second computes $f, Df, D^2f, \dots$ and tests B-linear dependence. Once an annihilator is found, apply the finite tests above and return their answer. A nonstable input has a failed moment, whereas a stable input is D-finite by Theorem 2.7; hence one branch terminates.

Finite B-linear dependence is decidable from the proper presentation: the transcendental generators are algebraically independent over B, and the full field is finite algebraic over their rational function field. Expand in an algebraic basis, clear denominators in the transcendental generators, and equate the finitely many monomial coefficients. This gives finite linear algebra over B. Thus an independent total test for D-finiteness is not assumed. Nor does the algorithm require effective versions of the existential bounds in Lemma 2.5; those bounds guarantee its termination. The elementary-integrability component is the usual finite-terms problem [24]; it is kept explicit as an input assumption rather than conflated with the new stability decision.

2.6 Polynomial weights and the remaining moment conjecture

The relative primitive-space bound also controls sparse sets of integrable moments. For an elementary function f over $C(x)$, put

$$I_f = \{p \in C[x] : pf \text{ has an elementary primitive}\}.$$

This is a C-vector space; it need not be an ideal of $C[x]$.

Theorem 2.12 (Polynomial-weight dichotomy)**.** *For every elementary function f over $C(x)$, exactly one of the following alternatives holds:*

1. *f is D-finite and I_f has finite codimension in $C[x]$;*
2. *f is not D-finite and I_f is finite-dimensional over C.*

In particular, infinitely many elementarily integrable monomial moments $x^n f$ force f to be D-finite.

Proof. The case $f = 0$ is immediate, with $I_f = C[x]$. The degree argument giving (2.9) applies to every nonzero operator $P \in C[x][D]$: its image $P(C[x])$ has finite codimension in $C[x]$. If $Lf = 0$ for some nonzero $L \in C[x][D]$, the Lagrange identity gives $L^*(C[x]) \subseteq I_f$. Thus D-finiteness implies the first alternative.

It remains to prove that infinite dimension of I_f forces D-finiteness. Induct on the transcendental length of an elementary tower containing f. At length zero, f is algebraic. At a positive layer (2.1), apply Lemma 2.6 to Bf. Its conclusion supplies a fixed finite algebraic extension F'/F, an elementary extension A of EF', and a fixed finite-dimensional B-space $W \subset A$ such that, for every $p \in I_f$,

$$pf = Dw_p + q_p, \qquad w_p \in W, \quad q_p \in F'.$$

Let d be the dimension of the image of W in the differential B-module A/F'. Choose $d+1$ polynomials in I_f that are linearly independent over C. Lemma 2.2, applied to their

corresponding primitive classes, gives a nonzero $L \in B[D]$ with $Lf \in F'$; if $f \in F'$ already, one can simply take $L = 1$. Multiplying L on the left by a common denominator preserves this conclusion and permits $L \in C[x][D]$. Put $h = Lf$ and $P = L^*$.

For every $q \in C[x]$, the Lagrange identity expresses $qh - fP(q)$ as the derivative of an elementary function. Therefore

$$I_h = P^{-1}(I_f). \tag{2.10}$$

The intersection $I_f \cap P(C[x])$ is infinite-dimensional, because $P(C[x])$ has finite codimension. Its inverse image under P is infinite-dimensional as well. Equation (2.10) therefore makes I_h infinite-dimensional. The field F' has smaller transcendental length, so induction makes h D-finite. Composing a nonzero annihilator of h with L gives a nonzero annihilator of f.

This proves the dichotomy. Finally, distinct monomials are linearly independent over C, so an infinite set of integrable monomial moments makes I_f infinite-dimensional. □

Chen's final conjecture in [20, Section 5] asks whether, for every elementary f, the unchanged set

$$S_f = \{n \geq 0 : x^n f \text{ is elementary integrable over } C(x)\}$$

is a Skolem set. We use the usual convention: a finite union of infinite arithmetic progressions together with a finite set. Theorem 2.12 proves the assertion for non-D-finite f, since S_f is then finite. The other case has the following precise reduction.

Proposition 2.13 (Holonomic obstruction sequences)**.** *If f is elementary and D-finite, then S_f is the simultaneous zero set of finitely many P-recursive sequences over C. Consequently, a Skolem theorem for all P-recursive sequences over C would imply Chen's full elementary-moment conjecture.*

Proof. The quotient $Q_f = C[x]/I_f$ is finite-dimensional by Theorem 2.12. Write $e_n = [x^n] \in Q_f$, choose a nonzero annihilator $L \in C[x][D]$ of f, and put $P = L^* = \sum_i b_i(x)D^i$. Since $P(C[x]) \subseteq I_f$, expansion of $P(x^n)$ gives, for all sufficiently large n,

$$\sum_j c_j(n)e_{n+j} = 0, \qquad c_j(z) = \sum_i [x^{i+j}]b_i(x)\,(z)_i. \tag{2.11}$$

Only finitely many j occur; coefficients with negative polynomial indices are understood to be zero. The polynomials c_j are not all zero, since the falling factorials $(z)_i$ are linearly independent and $P \neq 0$. Thus each coordinate sequence of e_n in a fixed C-basis of Q_f is P-recursive. These coordinates vanish simultaneously exactly when $x^n \in I_f$. If $Q_f = 0$, take the single identically zero sequence.

Finite intersections of Skolem sets are Skolem sets. The conditional conclusion follows, with the non-D-finite case supplied by Theorem 2.12. □

The remaining zero-set step is substantial. The finite-dimensional quotient Q_f does not in general carry multiplication by x, so (2.11) need not have constant coefficients. For example, take $f = e^{x^2}$ and $P = -D - 2x$. Then $e_0 \neq 0$, $e_1 = 0$, and

$$e_{n+1} = -\frac{n}{2}e_{n-1} \quad (n \geq 1).$$

In particular, $x \in I_f$ but $x^2 \notin I_f$. The Skolem–Mahler–Lech theorem for constant-coefficient recurrences therefore cannot be invoked solely from finite dimension of Q_f. Rubel's holonomic zero-set question and the special positive results of Bell, Burris and Yeats [27] concern precisely the kind of additional recurrence information needed here. The reduction above leaves Chen's full moment-set conjecture unresolved; it is a strengthening of the present structural argument, not a second named problem resolved by this section.

Audit boundary. The claimed mathematical conclusion is the complete characterization in Theorem 2.9, not merely a theorem for rational, logarithmic, or unramified inputs. The decisive additional steps are the all-place bounds before Riemann–Roch, the finite logarithmic-divisor space modulo the lower field, and the quotient-module argument. Independent reviews within this research session found no fatal gap in these steps. This is not external peer review, formal verification, or proof that no identical argument exists in the literature.

3 A Bell–cumulant criterion and functional digraphs

This section proves a criterion for coefficientwise Hankel total positivity of polynomial sequences defined by an exponential formula. Its application establishes the mathematical assertions in Sokal's Conjecture 6.9(b) and (d) [1], with their original indexing and normalization. The substantive step is a passage from positive semidefinite coefficient kernels to *all* Hankel minors, using auxiliary polynomial weights and Schur functions. The proof does not assume that a canonical production matrix is totally positive.

We use "totally positive" in the non-strict convention: every finite minor is nonnegative. A real sequence $(a_n)_{n\geq 0}$ is a Stieltjes moment sequence if $a_n = \int_{[0,\infty)} x^n \, d\rho(x)$ for a positive measure with all moments finite. The classical Stieltjes criterion identifies this property with total positivity of $[a_{i+j}]_{i,j\geq 0}$; equivalently, both $[a_{i+j}]$ and $[a_{i+j+1}]$ are positive semidefinite on every finite principal block. For polynomials in t, coefficientwise Hankel total positivity means that every minor of the Hankel matrix belongs to $\mathbb{R}_{\geq 0}[t]$.

3.1 The exact criterion

Let $(\kappa_n)_{n\geq 1}$ be an arbitrary real sequence, and define formal series and polynomials by

$$K(z) = \sum_{n\geq 1} \kappa_n \frac{z^n}{n!}, \qquad M_t(z) = \exp(tK(z)) = \sum_{n\geq 0} m_n(t) \frac{z^n}{n!}. \tag{3.1}$$

Thus $m_0(t) = 1$ and $m_n(t)$ is the complete exponential Bell polynomial evaluated at $t\kappa_1, \ldots, t\kappa_n$. No convergence of K is assumed.

Theorem 3.1 (Bell–cumulant criterion)**.** *For the sequence in* (3.1)*, the following statements are equivalent:*

1. *The shifted cumulant sequence $(\kappa_{n+1})_{n\geq 0}$ is a Stieltjes moment sequence.*
2. *The sequence $(m_n(t))_{n\geq 0}$ is coefficientwise Hankel-totally positive in t.*
3. *For every real $t > 0$, the numerical sequence $(m_n(t))_{n\geq 0}$ is a Stieltjes moment sequence.*

The necessity direction uses the standard lowest-coefficient argument; compare the related extraction in Zhu's Theorem 3.12(iii) [3]. The substantive direction proved here is sufficiency.

We first give two elementary positivity lemmas. Call a formal bivariate series $A(u,v) = \sum_{i,j\geq 0} a_{ij} u^i v^j$ a *positive semidefinite coefficient kernel* if every finite principal block of its coefficient matrix $[a_{ij}]$ is real positive semidefinite. This definition concerns coefficient matrices, rather than pointwise evaluation of A.

Lemma 3.2. *Positive semidefinite coefficient kernels are closed under addition, multiplication, and multiplication by nonnegative real scalars.*

Proof. Only multiplication requires explanation. Fix a truncation degree N. Write the required coefficient blocks of two kernels as Gram matrices, $a_{ij} = \langle v_i, v_j \rangle$ and $b_{ij} = \langle w_i, w_j \rangle$, for $0 \le i, j \le N$. For $0 \le n \le N$, set

$$q_n = \sum_{i=0}^{n} v_i \otimes w_{n-i}.$$

The coefficient of $u^n v^m$ in their product is $\langle q_n, q_m \rangle$. Its block through degree N is therefore positive semidefinite. Since N is arbitrary, this proves the assertion. All sums used here are finite. □

Lemma 3.3. *If $G_1, \dots, G_d$ are real positive semidefinite $r \times r$ matrices, then*

$$\det\left(\sum_{j=1}^{d} z_j G_j\right) \in \mathbb{R}_{\ge 0}[z_1, \dots, z_d].$$

Proof. Factor $G_j = V_j V_j^{\mathsf{T}}$ and concatenate the columns of the V_j into a matrix V. Then $\sum_j z_j G_j = V D V^{\mathsf{T}}$, where D is diagonal and each column belonging to V_j receives the diagonal entry z_j. Cauchy–Binet expands the determinant as a sum over r-element column sets S of

$$(\det V_S)^2 \prod_{c \in S} D_{cc}.$$

Every coefficient is a sum of nonnegative real squares. □

Proof of Theorem 3.1. We prove the substantive implication (1)⇒(2) first. Choose a positive measure ρ on $[0, \infty)$ such that

$$\kappa_{n+1} = \int x^n \, d\rho(x), \qquad n \ge 0. \tag{3.2}$$

Step 1: positive semidefinite coefficient kernels. For every integer $s \ge 1$, the coefficient matrix of $K^{(s)}(u+v)$ is

$$\left[\frac{\kappa_{i+j+s}}{i!j!}\right]_{i,j \ge 0}.$$

Its finite quadratic forms are integrals of squares multiplied by x^{s-1}, so it is positive semidefinite. The same is true of

$$C(u,v) = K(u+v) - K(u) - K(v) = \sum_{i,j \ge 1} \frac{\kappa_{i+j}}{i!j!} u^i v^j. \tag{3.3}$$

Indeed, its nonzero coefficient block has quadratic form

$$\sum_{i,j \ge 1} c_i c_j \frac{\kappa_{i+j}}{i!j!} = \int x \left(\sum_{i \ge 1} \frac{c_i x^{i-1}}{i!}\right)^2 d\rho(x) \ge 0,$$

where the c_i have finite support. This expression remains valid when ρ has an atom at zero.

For $s \ge 0$, put

$$Q_s(u,v;t) = \frac{M_t^{(s)}(u+v)}{M_t(u) M_t(v)}. \tag{3.4}$$

The quotients are well-defined formally because $M_t(0) = 1$. For $s \ge 1$, Bell differentiation gives

$$Q_s = e^{tC(u,v)} s! \sum_{\substack{\ell_1, \dots, \ell_s \ge 0 \\ \ell_1 + 2\ell_2 + \cdots + s\ell_s = s}} \prod_{j=1}^{s} \frac{t^{\ell_j} \left(K^{(j)}(u+v)\right)^{\ell_j}}{\ell_j! \, (j!)^{\ell_j}}, \tag{3.5}$$

and $Q_0 = e^{tC}$. Lemma 3.2 shows that every coefficient of t in each Q_s is a positive semidefinite coefficient kernel. Moreover, every finite coefficient block of Q_s is polynomial in t: C has positive degree in both u and v, while the Bell polynomial in (3.5) has finite degree in t.

Step 2: weighted leading determinants are Schur-positive. Fix $r \geq 1$ and $p \geq 1$. Introduce auxiliary variables $a = (a_1, \ldots, a_p)$ and the polynomial

$$P_a(x) = \prod_{\ell=1}^{p}(x + a_\ell) = \sum_{s=0}^{p} e_{p-s}(a)x^s,$$

where e_j denotes the elementary symmetric polynomial. Let L_t be the $\mathbb{R}[t]$-linear functional on $\mathbb{R}[t][x]$ determined by $L_t(x^n) = m_n(t)$. Consider

$$W_{r,p}(a;t) = \det\big[L_t(x^{i+j}P_a(x))\big]_{0\leq i,j<r}. \tag{3.6}$$

Its factorial-normalized matrix has generating kernel

$$\sum_{s=0}^{p} e_{p-s}(a)M_t^{(s)}(u+v) = M_t(u)M_t(v)\sum_{s=0}^{p} e_{p-s}(a)Q_s(u,v;t). \tag{3.7}$$

Let $G_s(t)$ be the $r \times r$ coefficient block of Q_s, and let $A_r(t)$ be the unit lower-triangular matrix

$$(A_r)_{ij} = [z^{i-j}]M_t(z) \quad (i \geq j), \qquad (A_r)_{ij} = 0 \quad (i < j).$$

Coefficient extraction in (3.7), followed by finite truncation, gives the exact congruence

$$\big[L_t(x^{i+j}P_a(x))\big]_{0\leq i,j<r} = D_r A_r \left(\sum_{s=0}^{p} e_{p-s}(a)G_s(t)\right) A_r^{\mathsf{T}} D_r, \qquad D_r = \operatorname{diag}(0!, 1!, \ldots, (r-1)!).$$

No terms outside the displayed finite blocks contribute: the factors A_r are lower triangular. Since $\det A_r = 1$, it follows that

$$W_{r,p}(a;t) = \left(\prod_{i=0}^{r-1}(i!)^2\right)\det\left(\sum_{s=0}^{p} e_{p-s}(a)G_s(t)\right). \tag{3.8}$$

Write $G_s(t) = \sum_k t^k G_{s,k}$. Each $G_{s,k}$ is a real positive semidefinite matrix, by Step 1, and the sum is finite. Apply Lemma 3.3 with independent weights for these matrices, and then substitute $t^k e_{p-s}(a)$ for the corresponding weight. Equation (3.8) expresses $W_{r,p}$ as a nonnegative real linear combination of terms

$$t^N e_{j_1}(a)\cdots e_{j_r}(a).$$

The Pieri rule, or the Littlewood–Richardson rule, implies that products of elementary symmetric polynomials are Schur-positive [7, Chapter I]. Consequently every coefficient of t in $W_{r,p}(a;t)$ has nonnegative coefficients when expanded in the Schur basis in the p auxiliary variables.

Step 3: extract every minor with consecutive initial row indices. For a partition λ of length at most r, padded with zeros, define

$$J_\lambda = (\lambda_r, \lambda_{r-1}+1, \ldots, \lambda_1 + r - 1), \qquad \delta = (0, 1, \ldots, r-1),$$

and write $D_{I,J}(t) = \det[m_{i_a+j_b}(t)]_{a,b=1}^{r}$ for increasing nonnegative index tuples I, J. With $\Delta(x) = \prod_{1\leq a<b\leq r}(x_b - x_a)$, the alternant formula for Schur functions and the formal Andréief identity give

$$D_{\delta,J_\lambda}(t) = \frac{1}{r!}L_t^{\otimes r}(\Delta(x)^2 s_\lambda(x)). \tag{3.9}$$

Here $L_t^{\otimes r}$ acts separately in the variables $x_1, \dots, x_r$. The identity is purely algebraic: expand both alternants, apply the functional term by term, and group the $r!$ equal contributions obtained by simultaneous permutation of the variables. In particular, no representing measure for L_t is being assumed.

The rectangular form of the dual Cauchy identity is

$$\prod_{i=1}^{r}\prod_{\ell=1}^{p}(x_i + a_\ell) = \sum_{\lambda \subseteq (p^r)} s_\lambda(x) s_{\widehat{\lambda}}(a), \qquad \widehat{\lambda} = (r - \lambda'_p, \dots, r - \lambda'_1), \tag{3.10}$$

where λ' is the conjugate partition [7, Chapter I]. For example, it follows from the ordinary dual Cauchy identity after replacing a_ℓ by a_ℓ^{-1} and multiplying by $\prod_\ell a_\ell^r$; the resulting identity is polynomial. Applying the formal Andréief identity to (3.6), then using (3.10) and (3.9), yields

$$W_{r,p}(a;t) = \sum_{\lambda \subseteq (p^r)} D_{\delta,J_\lambda}(t) s_{\widehat{\lambda}}(a). \tag{3.11}$$

The map $\lambda \mapsto \widehat{\lambda}$ is injective, and its images have length at most p. Their Schur polynomials are therefore linearly independent in p variables. Step 2 and (3.11) imply

$$D_{\delta,J_\lambda}(t) \in \mathbb{R}_{\geq 0}[t].$$

Every partition of length at most r occurs by taking $p \geq \max(1, \lambda_1)$.

Step 4: pass to arbitrary row and column index sets. Every increasing nonnegative tuple of length r is uniquely J_λ for some partition λ of length at most r. Thus for arbitrary $I = J_\lambda$ and $J = J_\mu$, the same alternant calculation gives

$$D_{I,J}(t) = \frac{1}{r!} L_t^{\otimes r}(\Delta(x)^2 s_\lambda(x) s_\mu(x)).$$

The Littlewood–Richardson identity

$$s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu, \qquad c_{\lambda\mu}^\nu \in \mathbb{Z}_{\geq 0},$$

holds after specialization to r variables, with terms of length greater than r vanishing [7, Chapter I]. It follows that

$$D_{I,J}(t) = \sum_{\ell(\nu) \leq r} c_{\lambda\mu}^\nu D_{\delta,J_\nu}(t) \in \mathbb{R}_{\geq 0}[t]. \tag{3.12}$$

This proves (2) for all minors, not just principal or solid minors.

To finish the equivalence, (2)⇒(3) follows by evaluation at $t > 0$ and the classical Stieltjes criterion. Suppose (3) holds. For arbitrary increasing nonnegative tuples $I = (i_a)$ and $J = (j_b)$ of length r, the shifted minor

$$\det[m_{i_a + j_b + 1}(t)]_{a,b=1}^r$$

is nonnegative for every $t > 0$, since it is a minor of the moment Hankel matrix, using rows I and columns $J + 1$. For $n \geq 1$, $m_n(t) = t\kappa_n + O(t^2)$ as a polynomial at zero. Dividing this minor by t^r and taking $t \downarrow 0$ proves

$$\det[\kappa_{i_a + j_b + 1}]_{a,b=1}^r \geq 0.$$

The shifted cumulant Hankel matrix is totally positive. The Stieltjes criterion now gives (1). □

Remark 3.4. The measure in the hypothesis of Theorem 3.1 is numerical. The proof does not assert that replacing the κ_n by arbitrary polynomials with coefficientwise Hankel positivity preserves the same conclusion jointly in all parameters. Its determinant lemma uses real positive semidefinite coefficient matrices. This distinction matters for refinements by additional combinatorial statistics.

3.2 An explicit measure for connected functional digraphs

Let $T(z)$ be the rooted-tree series, characterized by $T(0) = 0$ and $T(z) = ze^{T(z)}$, and put

$$K(z) = -\log(1 - T(z)) = \sum_{n\geq 1} \kappa_n \frac{z^n}{n!}. \tag{3.13}$$

The following proposition establishes the exact shifted-cumulant input needed above, as well as its factorial-normalized version.

Proposition 3.5. *Let W be the principal Lambert function. The function*

$$w(x) = \frac{1}{\pi} \operatorname{Arg}(1 + W(-1/x + i0)), \qquad 0 < x < e, \tag{3.14}$$

is a probability density on $(0, e)$ and satisfies $0 < w(x) < 1/2$. For the coefficients in (3.13),

$$\frac{\kappa_{n+1}}{(n+1)!} = \int_0^e x^n w(x)\, dx, \qquad n \geq 0. \tag{3.15}$$

Consequently both $(\kappa_{n+1}/(n+1)!)_{n\geq 0}$ and $(\kappa_{n+1})_{n\geq 0}$ are Stieltjes moment sequences.

Proof. The identity $T(-s) = -W(s)$ gives

$$h(s) = -K(-s) = \log(1 + W(s)), \qquad F(s) = \frac{h(s)}{s},$$

where the singularity of F at zero is removable and $F(0) = 1$. We derive its Stieltjes representation directly. We use the standard principal-branch properties of W, including its cut $(-\infty, -1/e]$ and its boundary parametrization [5].

For $t > 1/e$, there is a unique $v \in (0, \pi)$ such that

$$t = \frac{v}{\sin v} e^{-v\cot v}, \qquad W(-t + i0) = -v\cot v + iv. \tag{3.16}$$

The parametrization follows by imposing that We^W be negative real. For completeness, its t-coordinate increases strictly from $1/e$ to infinity: the derivative of its logarithm is $1/v - 2\cot v + v\csc^2 v$, whose product with $v\sin^2 v$ is $(\sin v - v\cos v)^2 + v^2\sin^2 v > 0$. Furthermore $1 - v\cot v > 0$ because $\sin v - v\cos v$ has derivative $v\sin v > 0$ and limit zero at the origin. Hence

$$\theta(t) := \operatorname{Arg}(1 + W(-t + i0)) \in (0, \pi/2),$$

with $\theta(1/e+) = \pi/2$ and $\theta(t) \to 0$ as $t \to \infty$.

The principal logarithm in h introduces no additional cut: W maps the upper half-plane into itself, while $1 + W(s) > 0$ for real $s > -1/e$. Thus F is holomorphic on $\mathbb{C} \setminus (-\infty, -1/e]$, after removing its singularity at zero, and is real symmetric. On the upper side of its cut,

$$\operatorname{Im} F(-t + i0) = -\frac{\theta(t)}{t}. \tag{3.17}$$

At the branch endpoint, the square-root expansion of $W+1$ shows that F has only a logarithmic singularity. At infinity, the principal-branch asymptotics $W(s) = \log s - \log\log s + o(1)$ give $F(s) = O(\log\log|s|/|s|)$ along a large keyhole contour. Accordingly the small endpoint-circle and large-circle terms in Cauchy's formula both vanish. Combining the two sides of the cut using (3.17) gives, for $s > 0$,

$$F(s) = \frac{1}{\pi} \int_{1/e}^{\infty} \frac{\theta(t)}{t(t+s)}\, dt = \int_0^e \frac{w(x)}{1 + sx}\, dx, \tag{3.18}$$

where the second equality is the substitution $x = 1/t$. The bounds on θ show that w is positive and bounded by $1/2$. Letting $s \downarrow 0$ in (3.18) gives $\int_0^e w(x)\,dx = F(0) = 1$.

Both sides of (3.18) are analytic for $|s| < 1/e$. The geometric series for $(1+sx)^{-1}$ converges uniformly on every smaller closed disk, uniformly for $x \in [0, e]$. Since

$$F(s) = \sum_{n\geq 0} (-1)^n \frac{\kappa_{n+1}}{(n+1)!} s^n,$$

comparison of Taylor coefficients proves (3.15).

Finally let X have density w and, independently, let G have the gamma density ge^{-g} on $(0, \infty)$. Since $\mathbb{E}[G^n] = \Gamma(n+2) = (n+1)!$, we obtain

$$\mathbb{E}[(XG)^n] = (n+1)! \int_0^e x^n w(x)\,dx = \kappa_{n+1}.$$

Thus the product XG supplies a positive representing measure for the unnormalized shifted cumulants as well. □

Remark 3.6. The existence assertion in Proposition 3.5 also follows from established complete-Bernstein theory. The principal function W is complete Bernstein [5]; so is $\log(1+s)$, and complete Bernstein functions are closed under composition. It follows that $h(s) = \log(1+W(s))$ is complete Bernstein and that $h(s)/s$ is Stieltjes [6]. The contour calculation above makes the representing density and normalization explicit. The analytic closure principle itself is an existing theorem, not a new ingredient claimed here.

3.3 The original functional-digraph assertions

A functional digraph on $[n]$ has exactly one outgoing edge at each vertex, with loops permitted. Let $\psi_{n,k}$ count such digraphs with k weakly connected components, and define

$$\Psi_n(y) = \sum_{k=0}^{n} \psi_{n,k} y^k, \qquad \Psi_0(y) = 1.$$

Each connected component is a directed cycle of rooted labelled trees. The labelled exponential formula therefore gives

$$\sum_{n\geq 0} \Psi_n(y) \frac{z^n}{n!} = \exp\left(y \sum_{\ell \geq 1} \frac{T(z)^\ell}{\ell} \right) = (1 - T(z))^{-y} = e^{yK(z)}, \qquad \kappa_n = \psi_{n,1}. \tag{3.19}$$

These are exactly the definitions used in Sokal's Conjecture 6.9 [1].

Corollary 3.7 (Sokal's Conjecture 6.9(b) and (d))**.** *Every finite minor of* $[\Psi_{i+j}(y)]_{i,j\geq 0}$ *has nonnegative coefficients in* y*. In addition,*

$$\left(\frac{\psi_{n+1,1}}{(n+1)!} \right)_{n\geq 0}$$

is a Stieltjes moment sequence, with the probability density (3.14) *as a representing measure.*

Proof. Proposition 3.5 gives the numerical Stieltjes property of $(\kappa_{n+1})_{n\geq 0}$. Apply Theorem 3.1 with $t = y$ and use (3.19). The normalized assertion is precisely (3.15) with $\kappa_n = \psi_{n,1}$. □

Zhu's Proposition 6.18 [2] already establishes the coefficient-triangle assertion 6.9(a) and the numerical shifted-moment assertion 6.9(c), and proves pointwise Stieltjes positivity of $\Psi_n(y)$. The argument here proves the full coefficientwise assertion (b), and supplies a direct measure for (d). It does not rely on the indexing of the cumulant summation displayed in that earlier proof. The four mathematical clauses of the original Conjecture 6.9 are thus accounted for, with the earlier results explicitly credited.

Remark 3.8. No joint coefficientwise assertion is made for the additional cyclic-vertex parameter in Sokal's Conjecture 6.10, or for the unnormalized ordered-forest sequence in Conjecture 6.7. Neither is obtained by treating the numerical cumulant hypothesis as a coefficientwise hypothesis in a second indeterminate.

3.4 An obstruction to the canonical production-matrix criterion

The following exact computation explains why the usual sufficient criterion based on the canonical exponential-Riordan production matrix does not directly prove Corollary 3.7.

Proposition 3.9. *The canonical production matrix of the coefficient triangle* $[\psi_{n,k}]_{n,k\ge 0}$ *has a negative entry:*

$$P_{17,1} = -1120228.$$

Proof. The column exponential generating functions are $K(z)^k/k!$. Differentiating them shows that the production entries are

$$P_{n,k} = \frac{n!}{(k-1)!}[w^{n-k+1}]A(w) \quad (k \ge 1), \qquad P_{n,0} = 0, \qquad A(w) = K'(K^{-1}(w)).$$

Indeed, expanding $A(K(z))K(z)^{k-1}/(k-1)!$ in the basis $K(z)^n/n!$ gives exactly these coefficients. For the present K, write $T = 1 - e^{-w}$ and $z = Te^{-T}$. Since $K'(z) = e^T/(1-T)^2$, this gives

$$A(w) = \exp(2w + 1 - e^{-w}). \tag{3.20}$$

Let $a_n = A^{(n)}(0)$, and set $\ell_1 = 3$ and $\ell_j = (-1)^{j+1}$ for $j \ge 2$. Differentiating $A' = A(2 + e^{-w})$ repeatedly gives the integer recurrence

$$a_0 = 1, \qquad a_{n+1} = \sum_{j=0}^{n} \binom{n}{j} \ell_{j+1} a_{n-j}.$$

It yields, by exact integer arithmetic,

$$\begin{aligned}(a_0, \dots, a_8) &= (1, 3, 8, 19, 41, 84, 171, 347, 690),\\ (a_9, \dots, a_{17}) &= (1385, 2825, 5438, 11077, 24535, 33720, 102623, 350605, -1120228).\end{aligned}$$

Finally $P_{17,1} = 17![w^{17}]A(w) = a_{17}$. □

Thus this particular production matrix is not even entrywise nonnegative. This observation excludes direct application of a sufficient criterion requiring its total positivity; it does not exclude other production matrices or alternative path representations.

4 The DeTemple quotient in Yang's monotonicity conjecture

Let $\psi = \Gamma'/\Gamma$ denote the digamma function and put

$$R(x) = \psi\Big(x + \tfrac{1}{2}\Big) - \log x, \qquad x > 0. \tag{4.1}$$

Yang [8, Conjecture 1(i)] conjectured that the quotient

$$H(x) = \frac{-24\left(x^4 - \frac{31}{336}\right) R(x) + x^2 - \frac{7}{40}}{24\left(x^2 + \frac{7}{40}\right) R(x) - 1} \tag{4.2}$$

is increasing on $(0, \infty)$. The same assertion appears as Conjecture 1(i) in the published article of Yang, Chu, and Zhang [9]. We prove it for the stated quotient on the entire stated domain.

Theorem 4.1. *The denominator in (4.2) is strictly positive, and $H'(x) > 0$ for every $x > 0$. Moreover,*

$$\frac{155}{294} < H(x) < \frac{11165}{8284}, \qquad x > 0, \tag{4.3}$$

and both bounds are sharp, being the limits at zero and infinity, respectively.

The proof identifies the derivative of a quotient of Stieltjes transforms with the deficit in the covariance form of the Cauchy–Schwarz inequality. We first establish this identity for an arbitrary positive measure.

Proposition 4.2 (A variance identity for a Stieltjes quotient)**.** *Let T be a nonnegative random variable with finite second moment whose support contains at least three points. Define*

$$m_1 = \mathbb{E}[T], \qquad m_2 = \mathbb{E}[T^2], \qquad F(s) = \mathbb{E}\left[\frac{1}{s+T}\right], \qquad s > 0,$$

and

$$D(s) = (s + m_1)F(s) - 1.$$

Then $D(s) > 0$, and the function

$$J(s) = \frac{-(s^2 - m_2)F(s) + s - m_1}{D(s)} \tag{4.4}$$

satisfies

$$J'(s) = \frac{\operatorname{Var}(T)\operatorname{Var}((s+T)^{-1}) - \operatorname{Cov}(T, (s+T)^{-1})^2}{D(s)^2} > 0. \tag{4.5}$$

Proof. Fix $s > 0$ and write $U = (s + T)^{-1}$. Since $TU = 1 - sU$, we have

$$D(s) = -\operatorname{Cov}(T, U). \tag{4.6}$$

If T' is an independent copy of T, with $U' = (s + T')^{-1}$, then

$$2\operatorname{Cov}(T, U) = \mathbb{E}[(T - T')(U - U')] < 0.$$

Indeed, $t \mapsto (s + t)^{-1}$ is strictly decreasing, and $\mathbb{P}(T \neq T') > 0$. All expectations are finite. This proves $D(s) > 0$.

Put $\sigma^2 = m_2 - m_1^2$. Algebraic rearrangement of (4.4) gives

$$J(s) = \frac{\sigma^2 F(s)}{D(s)} - s + m_1. \tag{4.7}$$

Differentiation under the expectation is justified locally in $s > 0$ by the bound $(s+T)^{-2} \leq s^{-2}$. Thus

$$F'(s) = -\mathbb{E}[U^2], \qquad D'(s) = F(s) + (s + m_1)F'(s).$$

It follows that

$$F'(s)D(s) - F(s)D'(s) = -F'(s) - F(s)^2 = \operatorname{Var}(U).$$

Differentiating (4.7) and using (4.6) proves the identity in (4.5).

Cauchy–Schwarz makes its numerator nonnegative. Equality would imply $U = aT + b$ almost surely for some real a, b. Consequently the support of T would lie among the zeros of

$$(s + t)(at + b) - 1.$$

This is a nonzero polynomial of degree at most two: it cannot vanish identically, as comparison of its coefficients immediately shows. Its zero set contains at most two points, contrary to the support hypothesis. The inequality is therefore strict. □

Lemma 4.3 (The positive measure for the DeTemple remainder)**.** *For $x > 0$,*

$$R(x) = \int_0^\infty \frac{du}{(x^2+u)(e^{2\pi\sqrt{u}}+1)}. \tag{4.8}$$

The measure

$$d\mu(u) = \frac{24\,du}{e^{2\pi\sqrt{u}}+1}, \qquad u > 0, \tag{4.9}$$

is a probability measure. Its first three moments are

$$m_1 = \frac{7}{40}, \qquad m_2 = \frac{31}{336}, \qquad m_3 = \frac{127}{1280}. \tag{4.10}$$

Proof. Binet's integral for the digamma function reads

$$\psi(x) = \log x - \frac{1}{2x} - 2\int_0^\infty \frac{t\,dt}{(x^2+t^2)(e^{2\pi t}-1)}.$$

Combining it with the duplication identity

$$\psi\Big(x+\tfrac{1}{2}\Big) = 2\psi(2x) - \psi(x) - 2\log 2$$

and changing the variable t to $2t$ in the integral arising from $\psi(2x)$ gives

$$\begin{aligned} R(x) &= 2\int_0^\infty \frac{t}{x^2+t^2}\left(\frac{1}{e^{2\pi t}-1} - \frac{2}{e^{4\pi t}-1}\right) dt \\ &= 2\int_0^\infty \frac{t\,dt}{(x^2+t^2)(e^{2\pi t}+1)}. \end{aligned}$$

The substitution $u = t^2$ proves (4.8).

For every integer $j \geq 0$, direct integration yields

$$\int_0^\infty u^j\,d\mu(u) = \frac{48\,\Gamma(2j+2)}{(2\pi)^{2j+2}}(1-2^{-2j-1})\zeta(2j+2). \tag{4.11}$$

For example, the identity follows by setting $v = 2\pi\sqrt{u}$ and expanding $(e^v+1)^{-1} = \sum_{\ell\geq 1}(-1)^{\ell-1}e^{-\ell v}$. The integrated series is absolutely summable for the powers v^{2j+1} occurring here, so termwise integration is justified. At $j = 0$ the expression in (4.11) is 1. At $j = 1, 2, 3$ the usual evaluations of the even zeta values give (4.10). □

Proof of Theorem 4.1. Let T have distribution (4.9). Its density is strictly positive throughout $(0,\infty)$, and it has moments of every order. Lemma 4.3 identifies its Stieltjes transform as

$$F(s) = 24R(\sqrt{s}).$$

With the moment values in (4.10), the denominator in (4.2) equals $D(x^2)$, and its numerator equals the numerator in (4.4) evaluated at $s = x^2$. Thus

$$H(x) = J(x^2), \qquad H'(x) = 2xJ'(x^2) > 0$$

by Proposition 4.2. The same proposition proves that the denominator is positive everywhere on the original domain.

It remains to evaluate the endpoints. Since ψ is continuous at $1/2$,

$$R(x) = -\log x + \psi(1/2) + o(1) \qquad (x \downarrow 0).$$

Consequently, dividing the numerator and denominator of $J(s)$ by $F(s)$ shows that

$$\lim_{x\downarrow 0} H(x) = \frac{m_2}{m_1} = \frac{155}{294}.$$

At infinity, the finite third moment justifies the Stieltjes expansion

$$F(s) = \frac{1}{s} - \frac{m_1}{s^2} + \frac{m_2}{s^3} - \frac{m_3}{s^4} + o(s^{-4}).$$

For clarity, this follows from finite geometric division of $(s+T)^{-1}$ and dominated convergence; convergence of an infinite asymptotic series is not being assumed. Substitution gives

$$D(s) = \frac{m_2 - m_1^2}{s^2} + o(s^{-2}), \qquad -(s^2 - m_2)F(s) + s - m_1 = \frac{m_3 - m_1 m_2}{s^2} + o(s^{-2}).$$

Therefore

$$\lim_{x\to\infty} H(x) = \frac{m_3 - m_1 m_2}{m_2 - m_1^2} = \frac{11165}{8284}.$$

Strict monotonicity now proves (4.3), including sharpness of both constants. □

Remark 4.4. The result proves the separately stated monotonicity assertion in Conjecture 1(i). The all-orders coefficient problem is treated separately in Section 5. The companion Conjecture 1(ii) is false: Kang's earlier preprint [28] supplies a strictly interior counterexample. It is not counted as a new result here. The integral representation and Cauchy–Schwarz inequality used here are classical; the quotient identity (4.5) supplies their application to the specific conjecture.

5 All orders in Yang's complete-monotonicity problem

Problem and result. Let

$$R(x) = \psi(x + \tfrac{1}{2}) - \log x, \qquad x > 0,$$

where $\psi = \Gamma'/\Gamma$. Problem 1 of Yang's 2014 preprint [8], repeated in the 2015 article of Yang, Chu, and Zhang [9], asks for the coefficients for which

$$\left(\sum_{j=0}^{n+1} a_j x^{2j}\right) R(x) - \sum_{j=0}^{n} b_j x^{2j}$$

is completely monotone and has a finite nonzero leading coefficient at order x^{-2n-4}. There is an unavoidable positive scaling freedom. The theorem below determines every solution and proves strict complete monotonicity at every order. The argument is presented as an internally audited complete-proof candidate. The bounded priority review is recorded in the status section; historical novelty is not certified.

Define the probability measure

$$d\mu(u) = \frac{24\,du}{e^{2\pi\sqrt{u}} + 1}, \qquad u > 0, \tag{5.1}$$

and its moments

$$m_j = \int u^j\, d\mu(u) = \frac{24(1 - 2^{-2j-1})|B_{2j+2}|}{2j+2}, \qquad j \geq 0. \tag{5.2}$$

Here B_j are the Bernoulli numbers; in particular $m_0 = 1$. Let π_N be the monic orthogonal polynomial of degree N for μ, and write

$$H_N = \int \pi_N(u)^2 \, d\mu(u) > 0, \qquad P_N(s) = (-1)^N \pi_N(-s) = \sum_{j=0}^{N} a_j s^j.$$

Thus $a_N = 1$. Put

$$Q_{N-1}(s) = \int \frac{P_N(s) - P_N(-u)}{s+u} \, d\mu(u). \tag{5.3}$$

The quotient in the integrand is a polynomial, so Q_{N-1} has degree at most $N-1$.

Theorem 5.1. *For every $N \geq 1$, the function*

$$E_N(x) = P_N(x^2)R(x) - \frac{1}{24} Q_{N-1}(x^2) \tag{5.4}$$

is strictly completely monotone on $(0,\infty)$: for every integer $k \geq 0$ and every $x > 0$,

$$(-1)^k E_N^{(k)}(x) > 0.$$

Moreover,

$$\lim_{x\to\infty} x^{2N+2} E_N(x) = \frac{H_N}{24}. \tag{5.5}$$

Every solution of the stated problem with $n = N-1$ is a positive constant multiple of (5.4).

In explicit finite-dimensional terms, if

$$\Delta_N = \det(m_{i+j})_{i,j=0}^{N-1}, \qquad \Delta_0 = 1,$$

then

$$\pi_N(u) = \frac{1}{\Delta_N} \det \begin{pmatrix} m_0 & m_1 & \cdots & m_N \\ m_1 & m_2 & \cdots & m_{N+1} \\ \vdots & \vdots & & \vdots \\ m_{N-1} & m_N & \cdots & m_{2N-1} \\ 1 & u & \cdots & u^N \end{pmatrix}, \qquad H_N = \frac{\Delta_{N+1}}{\Delta_N}. \tag{5.6}$$

The normalized coefficients in the question are the coefficients a_j of P_N, together with

$$b_k = \frac{1}{24} \sum_{j=k+1}^{N} a_j (-1)^{j-k-1} m_{j-k-1}, \qquad 0 \leq k < N. \tag{5.7}$$

The structural question behind the proof is when a Dirichlet average of an orthogonal-polynomial ensemble is a mixture of gamma laws of one fixed shape. Lemma 5.3 supplies a sufficient spectral condition, and the Yang weight satisfies it at every matrix size. This new auxiliary question is not counted as a pre-existing conjecture. The decisive positivity statement is obtained from a one-dimensional marginal of a positive Hermitian matrix. We give the matrix argument and its transform identity explicitly; no convergence of an infinite moment expansion is assumed.

5.1 A matrix marginal and gamma mixtures

For a positive function f and a positive-definite Hermitian matrix X, the notation $\det f(X)$ means the product of f evaluated at the eigenvalues of X.

Lemma 5.2 (A spectral product class)**.** *Suppose*

$$f(u) = Cu^{-\beta}e^{-c\sqrt{u}} \prod_{j\geq 1}(1+d_j u)^{-\gamma_j}, \qquad u > 0, \tag{5.8}$$

where $C > 0$, $\beta, c, \gamma_j \geq 0$, $d_j > 0$, and $\sum_j \gamma_j d_j < \infty$; finite products are allowed. If $A \geq 0$ is Hermitian, $e \neq 0$, and $A + see^* > 0$ for every $s > 0$, then

$$s \longmapsto \det f(A + see^*)$$

is completely monotone on $(0, \infty)$.

Proof. The matrix determinant lemma shows that $\det(I + d_j(A + see^*))$ is a positive constant times $1 + sd_j e^*(I + d_j A)^{-1}e$. Its negative power is completely monotone. Also $\det(A + see^*)$ is affine in s, with nonnegative coefficients and positive values for $s > 0$; its negative power has the same property. This assertion follows directly from the determinant lemma when $A > 0$, and by a limit from $A + \varepsilon I$ otherwise.

For the square-root factor, the scalar identity

$$\sqrt{x} = \frac{1}{2\pi}\int_0^\infty t^{-1/2}\log(1 + x/t)\,dt, \qquad x \geq 0,$$

first follows by differentiation for $x > 0$ and then by continuity at zero. Functional calculus and the determinant lemma give

$$\operatorname{Tr}\sqrt{A + see^*} - \operatorname{Tr}\sqrt{A} = \frac{1}{2\pi}\int_0^\infty t^{-1/2}\log(1 + sb(t))\,dt, \qquad b(t) = e^*(A + tI)^{-1}e \geq 0. \tag{5.9}$$

The integral converges, including for singular A, since $b(t) \leq \|e\|^2/t$: the resulting majorant is integrable both at zero and at infinity for each fixed $s > 0$. Approximate this positive integral by finite positive quadrature sums on compact subintervals of $(0, \infty)$ and then let those intervals increase. The corresponding exponentials are positive products of functions $(1 + sb)^{-a}$ with $a \geq 0$, hence are completely monotone. The class of completely monotone functions is closed under pointwise limits that are finite on $(0, \infty)$. Therefore $\exp(-c \operatorname{Tr}\sqrt{A + see^*})$ is completely monotone. Finally, the infinite product in (5.8) converges to a positive finite value on every positive-definite matrix, by $\sum_j \gamma_j d_j < \infty$. Apply the same pointwise-limit closure to its finite products and combine the factors. □

Lemma 5.3 (Gamma-mixture matrix marginal)**.** *Let $\alpha > 0$, let $r \geq 1$, and suppose a probability measure ν on $(0, \infty)$ has density*

$$w(u) = u^{\alpha-1}f(u) > 0$$

and moments of every order. Assume that $s \mapsto \det f(A + see^*)$ is completely monotone whenever $A \geq 0$ and $A + see^* > 0$ for $s > 0$. Let $(U_1, \ldots, U_r)$ have density proportional to

$$\prod_{i<j}(U_i - U_j)^2 \prod_{i=1}^r w(U_i)$$

on $(0, \infty)^r$. Independently let $(V_1, \ldots, V_r)$ have the Dirichlet$(1, \ldots, 1)$ distribution, and put $S = \sum_i V_i U_i$. Then S is a positive mixture of gamma laws with common shape $p = \alpha + r - 1$ and varying positive rates.

Proof. Consider the probability density proportional to

$$\det(X)^{\alpha-1}\det f(X)$$

on the cone of positive-definite complex Hermitian $r\times r$ matrices, with respect to Lebesgue measure. Its integral is finite because the spectral Jacobian is a constant times $\prod_{i<j}(U_i-U_j)^2$ and all the required moments are finite. The spectral change of variables also shows that its eigenvalues have the indicated density and that its eigenvectors are independently Haar unitary. The squared absolute values of the first row of a Haar unitary matrix have the Dirichlet$(1,\dots,1)$ distribution. For example, this follows by normalizing independent standard complex Gaussians, whose squared absolute values are independent exponentials. Consequently X_{11} has the law of S.

For $r>1$, use the Schur-complement parametrization

$$X=\begin{pmatrix} s & \sqrt{s}\,z^* \\ \sqrt{s}\,z & B+zz^* \end{pmatrix}, \qquad s>0, \quad B>0, \quad z\in\mathbb{C}^{r-1}. \tag{5.10}$$

It is bijective, $\det X = s\det B$, and its real Jacobian is s^{r-1}. Indeed the $2r-2$ real coordinates of the off-diagonal vector are scaled by $\sqrt{s}$, while translation of the lower block has Jacobian one. Furthermore $X=LL^*$, where

$$L=\begin{pmatrix} \sqrt{s} & 0 \\ z & B^{1/2} \end{pmatrix}.$$

Thus X has the same eigenvalues as

$$L^*L = A + se_1e_1^*, \qquad A=\begin{pmatrix} z^*z & z^*B^{1/2} \\ B^{1/2}z & B \end{pmatrix}\geq 0.$$

The density of S consequently has the form

$$p_S(s)=s^{p-1}G(s), \qquad G(s)=C_r\int_{B>0}\int_{\mathbb{C}^{r-1}}(\det B)^{\alpha-1}\det f(A+se_1e_1^*)\,dz\,dB. \tag{5.11}$$

Each integrand is completely monotone in s. Positive integration preserves complete monotonicity whenever the result is finite. Here finiteness holds for every $s>0$: the integral is nonincreasing, so an infinite value at any $s_0>0$ would contradict $\int_0^\infty s^{p-1}G(s)\,ds=1$. This argument can be formalized by exhausting the parameter domain with compact sets and applying pointwise-limit closure to the finite integrals. The case $r=1$ is the same calculation without B,z.

By Bernstein's theorem there is a positive measure ρ on $[0,\infty)$ such that $G(s)=\int e^{-\lambda s}\rho(d\lambda)$. Tonelli's theorem and normalization yield

$$1=\Gamma(p)\int_{(0,\infty)}\lambda^{-p}\rho(d\lambda), \qquad \rho(\{0\})=0.$$

Hence $\tau(d\lambda)=\Gamma(p)\lambda^{-p}\rho(d\lambda)$ is a probability measure and

$$p_S(s)=\int_{(0,\infty)}\frac{\lambda^p}{\Gamma(p)}s^{p-1}e^{-\lambda s}\,\tau(d\lambda),$$

which is the claimed gamma mixture. □

5.2 An exact orthogonal-polynomial transform

Lemma 5.4 (Divided-difference identity)**.** *Let ν be a probability measure on $(0,\infty)$ with all moments and infinite support. Let π_N be its monic orthogonal polynomial and $H_N = \int \pi_N^2\, d\nu$. Take $r = N+1$ and form U, V, S from ν as in Lemma 5.3. For a C^N function F for which the following integrals converge,*

$$\int \pi_N(u) F(u)\, d\nu(u) = \frac{H_N}{N!}\, \mathbb{E}[F^{(N)}(S)]. \tag{5.12}$$

In particular, this identity applies to $F(u) = \sin(t\sqrt{u})/\sqrt{u}$ for every fixed $t > 0$.

Proof. Write $Z_r = \int \prod_{i<j}(u_i - u_j)^2 \prod_i d\nu(u_i)$. Expanding the Vandermonde determinants and using orthogonality gives $Z_r = r! \prod_{j=0}^{r-1} H_j$. Heine's formula for π_N, followed by symmetrization in the $N+1$ integration variables, then gives

$$\int \pi_N(u) F(u)\, d\nu(u) = H_N\, \mathbb{E}\big[[U_1, \dots, U_{N+1}]F\big].$$

Explicitly, the symmetrized numerator is the full Vandermonde square multiplied by the divided difference

$$\sum_i \frac{F(U_i)}{\prod_{j\neq i}(U_i - U_j)}.$$

The normalizing factor is $Z_{N+1}/((N+1)N! \prod_{j<N} H_j) = H_N$. The Hermite–Genocchi formula gives

$$[U_1, \dots, U_{N+1}]F = \frac{1}{N!}\mathbb{E}\left[F^{(N)}\left(\sum_i V_i U_i \right) \middle| U \right],$$

proving (5.12). For the stated sine function,

$$F^{(N)}(u) = \frac{(-1)^N t^{2N+1}}{4^N (3/2)_N}\, {}_0F_1\left(N + \tfrac{3}{2}; -\frac{t^2 u}{4} \right). \tag{5.13}$$

The derivative formula follows from the entire power series of the single-variable function F, not from any moment expansion of ν. The bound $|{}_0F_1(p; -v)| \le 1$ for $p \ge \frac{1}{2}$, $v \ge 0$, follows from its cosine integral against a beta probability measure (with the endpoint $p = \frac{1}{2}$ giving a cosine). Thus F and $F^{(N)}$ are bounded here. The Heine integrals converge absolutely by the finite moments, so all symmetrizations and integrations above are justified. □

Lemma 5.5 (Yang kernel positivity)**.** *Let $q(t) = 1/t - 1/(2\sinh(t/2))$ and $h_N(t) = P_N(D_t^2)q(t)$, with P_N defined from (5.1). There is a probability measure τ_N on $(0,\infty)$ such that*

$$h_N(t) = \frac{H_N}{24(2N+1)!}\, t^{2N+1} \int_{(0,\infty)} e^{-t^2/(4\lambda)}\, \tau_N(d\lambda), \qquad t > 0. \tag{5.14}$$

In particular $h_N(t) > 0$, and $h_N(\sqrt{z})/z^{N+1/2}$ is completely monotone in $z > 0$.

Proof. Binet's digamma integral and the duplication formula give

$$R(x) = 2\int_0^\infty \frac{t\, dt}{(x^2+t^2)(e^{2\pi t}+1)} = \frac{1}{24}\int_0^\infty \frac{d\mu(u)}{x^2+u}. \tag{5.15}$$

Indeed, subtracting Binet's formula at x from twice its value at $2x$, and then using $\psi(x+\frac{1}{2}) = 2\psi(2x) - \psi(x) - 2\log 2$, replaces $(e^{2\pi t}-1)^{-1}$ by $(e^{2\pi t}-1)^{-1} - 2(e^{4\pi t}-1)^{-1} = (e^{2\pi t}+1)^{-1}$.

The ordinary digamma integral also gives $R(x) = \int_0^\infty e^{-xt} q(t)\, dt$. Since the Laplace transform of $\sin(t\sqrt{u})/\sqrt{u}$ is $(x^2+u)^{-1}$, uniqueness of Laplace transforms yields

$$q(t) = \frac{1}{24} \int \frac{\sin(t\sqrt{u})}{\sqrt{u}}\, d\mu(u).$$

Fubini is valid, for example by $|\sin(t\sqrt{u})/\sqrt{u}| \le t$; both sides are continuous. Differentiation under the integral is justified to every finite order by the moments of μ and gives

$$h_N(t) = \frac{(-1)^N}{24} \int \pi_N(u) \frac{\sin(t\sqrt{u})}{\sqrt{u}}\, d\mu(u).$$

Apply Lemma 5.3 with $\alpha = 3/2$ and

$$f(u) = \frac{24u^{-1/2}}{e^{2\pi\sqrt{u}}+1} = 12u^{-1/2}e^{-\pi\sqrt{u}} \prod_{j\ge1} \left(1 + \frac{4u}{(2j-1)^2}\right)^{-1}.$$

The last equality is Euler's product for $\cosh(\pi\sqrt{u})$. The reciprocal-square coefficients are summable, so this function satisfies Lemma 5.2. Thus, for $r = N+1$, the random variable S in Lemma 5.4 is a mixture of gamma laws with shape $p = N + 3/2$ and positive rate distribution τ_N. Equations (5.12) and (5.13) imply

$$\frac{h_N(t)}{t^{2N+1}} = \frac{H_N}{24\, 4^N N!(3/2)_N} \mathbb{E}\left[{}_0F_1\left(p; -\frac{t^2 S}{4}\right)\right].$$

For each fixed $\lambda > 0$, integration against the gamma density of shape p and rate λ gives

$$\int_0^\infty {}_0F_1(p; -zs/4) \frac{\lambda^p}{\Gamma(p)} s^{p-1} e^{-\lambda s}\, ds = e^{-z/(4\lambda)}, \qquad z \ge 0.$$

This identity follows by termwise integration of the hypergeometric series: the series of integrals of absolute values is $\exp(z/(4\lambda)) < \infty$ for the fixed λ. The subsequent integration over τ_N is justified by the bound $|{}_0F_1(p; -v)| \le 1$; no exponential moment of τ_N is used. Finally $4^N N!(3/2)_N = (2N+1)!$, proving (5.14). Strict positivity follows because τ_N is a probability measure on positive finite rates. □

5.3 Coefficients, complete monotonicity, and uniqueness

Proof of Theorem 5.1. The measure (5.1) is positive on $(0,\infty)$ and has all moments. Substituting $v = 2\pi\sqrt{u}$ and expanding $(e^v+1)^{-1}$ gives

$$m_j = \frac{48\Gamma(2j+2)}{(2\pi)^{2j+2}}(1 - 2^{-2j-1})\zeta(2j+2),$$

which is (5.2), including $m_0 = 1$. All Hankel determinants Δ_N are positive by positive definiteness of the moment form. The standard Gram determinant construction gives (5.6); it may also be obtained directly by solving the N orthogonality equations. Expanding the polynomial quotient in (5.3) proves (5.7).

The sine representation of q gives

$$q^{(2j)}(0) = 0, \qquad q^{(2j+1)}(0) = \frac{(-1)^j m_j}{24}.$$

Its derivatives of every fixed order are bounded, by the finite moments and $\int u^{-1/2}\, d\mu(u) < \infty$. Repeated integration by parts therefore shows

$$\int_0^\infty e^{-xt} q^{(2j)}(t)\, dt = x^{2j} R(x) - \frac{1}{24} \sum_{\ell=0}^{j-1} (-1)^\ell m_\ell x^{2j-2-2\ell}.$$

Combining these identities with the coefficients of P_N yields

$$E_N(x) = \int_0^\infty e^{-xt} h_N(t)\, dt. \tag{5.16}$$

By Lemma 5.5,

$$0 < h_N(t) \le \frac{H_N}{24(2N+1)!} t^{2N+1}, \qquad t > 0.$$

Consequently all derivatives of (5.16) can be taken under the integral, and they have the asserted strict signs. Also, dominated convergence in (5.14) gives

$$\lim_{t\downarrow 0} \frac{h_N(t)}{t^{2N+1}} = \frac{H_N}{24(2N+1)!}.$$

Changing variables $t = y/x$ in (5.16) and applying the same polynomial majorant proves (5.5).

For uniqueness, let A and B be real polynomials of degrees at most N and $N-1$, respectively, such that $A(x^2)R(x) - B(x^2)$ has a finite nonzero limit after multiplication by x^{2N+2}. Set $s = x^2$ and

$$Q_A(s) = \int \frac{A(s) - A(-u)}{s+u}\, d\mu(u).$$

Polynomial division in (5.15) shows that the polynomial part must vanish, so $B = Q_A/24$, and the remainder is

$$\frac{1}{24} \int \frac{A(-u)}{s+u}\, d\mu(u).$$

Finite geometric division, with a remainder controlled by the available moments, now forces

$$\int u^j A(-u)\, d\mu(u) = 0, \qquad 0 \le j < N.$$

The space of polynomials of degree at most N orthogonal to all lower-degree polynomials is one-dimensional. Therefore $A(s) = cP_N(s)$ and $B(s) = cQ_{N-1}(s)/24$ for a nonzero real c. Complete monotonicity implies nonnegativity of the function; its nonzero asymptotic coefficient is $cH_N/24$, so necessarily $c > 0$. Conversely, every such positive multiple satisfies all requirements by the result just proved. □

6 An interior counterexample to the Ferreira–Simon Wright conjecture

For $\rho > -1$, define the Wright function by

$$\phi(\rho, \beta, z) = \sum_{n=0}^{\infty} \frac{z^n}{n!\,\Gamma(\beta + \rho n)}.$$

The reciprocal Gamma function is understood at its zeros, so that this definition also covers nonpositive integral values of $\beta + \rho n$. For $0 < \alpha < 1$ and $\beta \ge 0$, the function

$$\varphi_{\alpha,\beta}(x) = \Gamma(\alpha+\beta)\phi(-\alpha, \beta, -x), \qquad x > 0,$$

is the probability density considered by Ferreira and Simon [10].

For $0 < \beta < 1$, let $\alpha_*(\beta)$ denote the unique solution in $(0,1)$ of

$$\frac{1}{\Gamma(\beta-\alpha)^2} = \frac{1}{\Gamma(\beta)\Gamma(\beta-2\alpha)}. \tag{6.1}$$

Set $\alpha_*(0) = 1$, as in the original paper. Section 6.4 of [10] conjectures the equivalence between $\alpha \le \alpha_*(\beta)$, log-concavity of this density, and convexity on $\mathbb{R}_+$ of the reciprocal Mittag–Leffler function $1/E_{\alpha,\alpha+\beta}(-t)$. In particular, the conjecture asserts

$$\alpha \le \alpha_*(\beta) \quad \Longrightarrow \quad \varphi_{\alpha,\beta} \text{ is log-concave.} \tag{6.2}$$

We disprove this implication within its original parameter domain.

The threshold test established in the same source will be used only in the region $0 < \beta < 2\alpha - 1 < 1$, where

$$\alpha > \alpha_*(\beta) \quad \Longleftrightarrow \quad R(\alpha,\beta) := \frac{\Gamma(\beta)\Gamma(\beta-2\alpha)}{\Gamma(\beta-\alpha)^2} < 1. \tag{6.3}$$

Consequently $R(\alpha,\beta) > 1$ in this region implies $\alpha < \alpha_*(\beta)$: the strict reverse side is excluded by (6.3), and equality is excluded by (6.1).

6.1 A limiting profile at the boundary

Theorem 6.1. *For all sufficiently small $\varepsilon > 0$, the density $\varphi_{1-\varepsilon,0}$ is strictly logarithmically convex at $x = 1/2$. More generally, if K is a compact subinterval of $(\sqrt{2}-1, 1)$, then*

$$\bigl(\log \varphi_{1-\varepsilon,0}\bigr)''(x) > 0 \qquad (x \in K)$$

for every sufficiently small $\varepsilon > 0$. Thus (6.2) is false.

Proof. Write $f_\varepsilon(x) = \phi(-1+\varepsilon, 0, -x)$. Euler reflection, including its continuous interpretation at Gamma poles, yields

$$f_\varepsilon(x) = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\Gamma(1+(1-\varepsilon)n)}{n!} \sin(\pi\varepsilon n) x^n. \tag{6.4}$$

Indeed, $1/\Gamma(-u) = -\Gamma(1+u)\sin(\pi u)/\pi$ and $\sin(\pi(1-\varepsilon)n) = (-1)^{n+1}\sin(\pi\varepsilon n)$. The $n = 0$ coefficient vanishes because $1/\Gamma(0) = 0$.

Log-convexity of Gamma on $[1, n+1]$ gives

$$0 < \Gamma(1+(1-\varepsilon)n) \le n!.$$

The coefficient of x^n in $f_\varepsilon/\varepsilon$ therefore has absolute value at most n, and it converges to n as $\varepsilon \downarrow 0$. On every interval $|x| \le r < 1$, this bound also gives a summable majorant for any fixed number of derivative series. Thus, for every fixed nonnegative integer k,

$$\frac{f_\varepsilon(x)}{\varepsilon} \longrightarrow h(x) := \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2} \quad \text{in } C^k \text{ on compact subintervals of } (-1,1). \tag{6.5}$$

In particular,

$$\frac{(f_\varepsilon')^2 - f_\varepsilon f_\varepsilon''}{\varepsilon^2} \longrightarrow (h')^2 - hh'' = \frac{1-2x-x^2}{(1-x)^6} \tag{6.6}$$

locally uniformly. The last expression equals -16 at $x = 1/2$ and is strictly negative for $\sqrt{2}-1 < x < 1$. The limiting profile is strictly positive there, so (6.5) also ensures positivity of f_ε on the compact intervals under consideration. Since

$$(\log f_\varepsilon)'' = -\frac{(f_\varepsilon')^2 - f_\varepsilon f_\varepsilon''}{f_\varepsilon^2},$$

the claimed strict logarithmic convexity follows. Multiplication by $\Gamma(1-\varepsilon) > 0$ does not change logarithmic curvature. Finally, $1 - \varepsilon < 1 = \alpha_*(0)$, proving the contradiction to (6.2). □

6.2 A completely explicit interior witness

The preceding limiting argument already disproves the conjecture. The next theorem supplies rational parameters with $\beta > 0$, using only inequalities and exact rational arithmetic.

Theorem 6.2. *Let*

$$\varepsilon = \frac{1}{10000}, \qquad \alpha = \frac{9999}{10000}, \qquad \beta = \frac{1}{100000000} = \varepsilon^2.$$

Then $\alpha < \alpha_*(\beta)$, but

$$\bigl(\log \varphi_{\alpha,\beta}\bigr)''(1/2) > 0.$$

In particular, these parameters give a counterexample strictly inside the original admissible region.

Proof. We first record the elementary Gamma bounds used below. Let $\psi = \Gamma'/\Gamma$ and let γ be Euler's constant. Monotonicity of ψ, together with $\psi(1) = -\gamma$, $0 < \gamma < 1$, and $\psi(n+1) = H_n - \gamma < n$, gives

$$-1 < \psi(t) < n \qquad (1 \le t \le n+1,\ n \ge 1). \tag{6.7}$$

Here $H_n = \sum_{j=1}^{n} 1/j$. Also, for $0 \le u \le 1$,

$$\Gamma(1+u) \le 1, \qquad \Gamma(1+u) \ge e^{-u} \ge 1 - u. \tag{6.8}$$

The upper bound follows from log-convexity and $\Gamma(1) = \Gamma(2) = 1$; the lower bound follows by integrating $\psi \ge -1$.

Put

$$f_{\alpha,b}(x) = \phi(-\alpha, b, -x), \qquad g_0(x) = \frac{f_{\alpha,0}(x)}{\varepsilon}, \qquad g(x) = \frac{f_{\alpha,\beta}(x)}{\varepsilon}.$$

By (6.4),

$$g_0(x) = \sum_{n=1}^{\infty} A_n x^n, \qquad A_n = R_n \frac{\sin(\pi\varepsilon n)}{\pi\varepsilon}, \qquad R_n = \frac{\Gamma(n+1-\varepsilon n)}{n!}.$$

Log-convexity gives $0 < R_n \le 1$. Furthermore,

$$\log R_n = -\int_{n+1-\varepsilon n}^{n+1} \psi(t)\,dt \ge -\varepsilon n^2$$

by (6.7), so $1 - R_n \le \varepsilon n^2$. The inequality $|\sin z - z| \le |z|^3/6$ holds for every real z. Hence

$$\begin{aligned} |A_n - n| &\le n(1 - R_n) + nR_n \left| \frac{\sin(\pi\varepsilon n)}{\pi\varepsilon n} - 1 \right| \\ &\le \varepsilon n^3 + \frac{\pi^2}{6}\varepsilon^2 n^3 \le 3\varepsilon n^3. \end{aligned} \tag{6.9}$$

The last inequality uses $\pi^2/6 < 2$ and $0 < \varepsilon < 1$.

At $x = 1/2$, summing (6.9) and its first two derivative majorants gives

$$|g_0 - h| \le 78\varepsilon, \qquad |g_0' - h'| \le 900\varepsilon, \qquad |g_0'' - h''| \le 11184\varepsilon. \tag{6.10}$$

For completeness, the geometric-power sums used here are

$$\sum_{n\ge1} \frac{n^3}{2^n} = 26, \qquad \sum_{n\ge1} \frac{n^4}{2^n} = 150, \qquad \sum_{n\ge1} \frac{n^5}{2^n} = 1082.$$

They follow by repeatedly applying $x\,d/dx$ to $x/(1-x)$. The second-derivative constant in (6.10) is $12(1082-150)=11184$.

We next control the change in b. Write

$$c_n(b)=\frac{(-1)^n}{n!\,\Gamma(b-\alpha n)}.$$

For $n\ge 1$ and $0\le b\le \alpha/2$, reflection gives

$$c_n(b)=\frac{(-1)^n\Gamma(1+\alpha n-b)\sin(\pi(b-\alpha n))}{\pi n!}.$$

The Gamma argument lies in $[1,n+1]$, where its ratio to $n!$ is at most one and $|\psi|\le n$. Differentiation with respect to b therefore yields

$$|c_n'(b)|\le n/\pi+1\le n+1\qquad (n\ge 1). \tag{6.11}$$

For $n=0$, recurrence gives

$$c_0(b)=\frac{b}{\Gamma(1+b)},\qquad c_0'(b)=\frac{1-b\psi(1+b)}{\Gamma(1+b)}.$$

When $0\le b\le 1/2$, (6.7) with $n=1$ shows that the numerator is positive and at most $1+b$. By (6.8), $|c_0'(b)|\le (1+b)/(1-b)\le 3$.

Summing these coefficient bounds at $x=1/2$ gives

$$\begin{aligned}|f_{\alpha,\beta}-f_{\alpha,0}|&\le 6\beta,\\ |f'_{\alpha,\beta}-f'_{\alpha,0}|&\le 16\beta,\\ |f''_{\alpha,\beta}-f''_{\alpha,0}|&\le 96\beta.\end{aligned} \tag{6.12}$$

Indeed,

$$3+\sum_{n\ge1}\frac{n+1}{2^n}=6,\qquad 2\sum_{n\ge1}\frac{n(n+1)}{2^n}=16,\qquad 4\sum_{n\ge1}\frac{n^3-n}{2^n}=96.$$

The convergent majorants justify all differentiations and summations.

Combining (6.10) and (6.12), using $\beta/\varepsilon=\varepsilon$, we obtain at $x=1/2$

$$|g-h|\le E_0:=\frac{21}{2500},\qquad |g'-h'|\le E_1:=\frac{229}{2500},\qquad |g''-h''|\le E_2:=\frac{141}{125}.$$

Since $(h,h',h'')=(2,12,80)$ at this point, direct expansion of $D_g=(g')^2-gg''$ gives

$$\begin{aligned}|D_g+16|&\le 24E_1+E_1^2+2E_2+80E_0+E_0E_2\\ &=\frac{32151661}{6250000}<6.\end{aligned}$$

Thus $D_g<-10$, while $g\ge 2-E_0>0$. This proves $(\log g)''(1/2)>0$ and hence the same strict inequality for $\varphi_{\alpha,\beta}$.

It remains to verify the side of the conjectured threshold. Put $a=\varepsilon+\beta$ and $b=2\varepsilon+\beta$. Gamma recurrence gives the exact identity

$$R(\alpha,\beta)=\frac{\Gamma(1+\beta)\Gamma(1+b)}{\Gamma(1+a)^2}\,\frac{a^2(1-a)^2}{\beta b(1-b)(2-b)}. \tag{6.13}$$

Here $a,b<1/2$. The first factor is at least $1/4$ by (6.8). Also, $a\ge\varepsilon$, $1-a>1/2$, $b\le 3\varepsilon$, $1-b<1$, and $2-b<2$, with every denominator factor positive. Consequently

$$R(\alpha,\beta)\ge\frac{\varepsilon}{96\beta}=\frac{625}{6}>100>1.$$

The parameters satisfy $0<\beta<2\alpha-1<1$, so (6.3) and (6.1) imply $\alpha<\alpha_*(\beta)$, as required. □

Both inequalities in Theorem 6.2 are strict. The derivative-series majorants above also establish joint continuity in (α, β) near this pair. The Gamma ratio is continuous there as well. Hence the counterexamples include a nonempty open subset of the two-dimensional parameter domain, rather than just the boundary family of Theorem 6.1.

6.3 Scope and priority boundary

The result is a negative answer to the unchanged universal implication (6.2), and therefore to the proposed three-way equivalence. It does not classify the full log-concavity region and does not separately decide equivalence between the log-concavity and reciprocal-convexity properties. The boundary family, the explicit interior pair, and its open neighborhood address one conjecture and are counted as one result.

The argument uses classical reflection, Gamma inequalities, coefficient estimates, and a curvature determinant. No historically new general principle is claimed. A bounded literature check through 3 November 2026 found no identical resolution; this included the original paper, author publication lists, and the later preprint *Mittag–Leffler functions and convex ordering* (arXiv:2512.04940v1). The absence of a result in that search does not prove priority. The explicit proof above is intended to make the mathematical assertion independently checkable while keeping that historical limitation separate.

7 A negative answer to the Yano–Yano–Yor negative-power question

For $\alpha > 1$, let C_α have the symmetric α-Cauchy density

$$f_{C_\alpha}(x) = \frac{\alpha \sin(\pi/\alpha)}{2\pi} \frac{1}{1+|x|^\alpha}, \qquad x \in \mathbb{R}. \tag{7.1}$$

Yano, Yano and Yor [11, Remark 2.9(iii)] asked whether $|C_\alpha|^{-p}$ is self-decomposable, or at least infinitely divisible, for $p > 0$. We answer this universal question negatively. The counterexample lies in $1 < \alpha < 2$, so it also applies if that smaller range motivated the question.

Theorem 7.1. *The random variable*

$$Y = |C_{4/3}|^{-1/12}$$

is not infinitely divisible, and hence is not self-decomposable. Its density and an independent-gamma representation are

$$f_Y(y) = \frac{8\sqrt{2}}{\pi} \frac{y^3}{1+y^{16}}, \qquad y > 0, \qquad Y \overset{d}{=} \left(\frac{G_{1/4}}{G_{3/4}}\right)^{1/16}, \tag{7.2}$$

where $G_{1/4}$ and $G_{3/4}$ are independent gamma random variables with the indicated shapes and unit scale.

We give a proof using only three moments. In particular, no numerical approximation is needed to certify the counterexample.

Lemma 7.2. *If a nonnegative infinitely divisible random variable V has finite third moment, then*

$$\kappa_3(V) := \mathbb{E}[V^3] - 3\mathbb{E}[V]\mathbb{E}[V^2] + 2\mathbb{E}[V]^3 \geq 0.$$

Proof. For each positive integer n, write V in distribution as a sum of n independent copies of a convolution root Z_n. The essential lower endpoint of the support of such a sum is n times the essential lower endpoint for a summand. An unbounded-below root would give an unbounded-below sum. Since V is nonnegative, every Z_n is therefore nonnegative. Its third moment is finite, because each nonnegative summand is bounded by their sum.

Put $\mu = \mathbb{E}[V]$ and $v = \mathrm{Var}(V)$. Independence gives

$$\mathbb{E}[Z_n] = \frac{\mu}{n}, \qquad \mathbb{E}[Z_n^2] = \frac{v}{n} + \frac{\mu^2}{n^2}.$$

Additivity of the third cumulant and $\mathbb{E}[Z_n^3] \geq 0$ imply

$$\kappa_3(V) = n\kappa_3(Z_n) \geq -\frac{3\mu v}{n} - \frac{\mu^3}{n^2}.$$

Let n tend to infinity. □

Lemma 7.3. *Let $1 < \alpha < 2$ and $0 < p < 1/3$, and put*

$$\theta = \frac{\pi}{\alpha}, \qquad h = \frac{\pi p}{\alpha}, \qquad a = \cot\theta, \qquad t = \tan h.$$

Define

$$\begin{aligned} B_1 &= 1 - at, \\ B_2 &= 1 - t^2 - 2at, \\ B_3 &= 1 - 3t^2 - at(3 - t^2). \end{aligned}$$

These three numbers are strictly positive, and

$$\kappa_3(|C_\alpha|^{-p}) = \frac{2t^3(1+a^2)(1+t^2)^{3/2}\big[a + (a^2+3)t - at^2\big]}{B_1^3 B_2 B_3}. \tag{7.3}$$

Proof. For $j = 1, 2, 3$, the condition $jp < 1$ and the beta integral give

$$m_j := \mathbb{E}[|C_\alpha|^{-jp}] = \frac{\alpha \sin\theta}{\pi} \int_0^\infty \frac{x^{-jp}}{1 + x^\alpha}\, dx = \frac{\sin\theta}{\sin(\theta - jh)}. \tag{7.4}$$

Indeed, after substituting $u = x^\alpha$, the beta-integral parameter is $(1 - jp)/\alpha \in (0, 1)$. Also $0 < h < \theta/3 < \pi/3$, so $\cos h > 0$. The angle-subtraction identities show that

$$B_j = \frac{\sin(\theta - jh)}{\sin\theta \cos^j h} > 0, \qquad m_j = \frac{(1+t^2)^{j/2}}{B_j} \quad (j = 1, 2, 3).$$

Consequently

$$m_3 - 3m_1m_2 + 2m_1^3 = \frac{(1+t^2)^{3/2}}{B_1^3 B_2 B_3}(B_1^3 B_2 - 3B_1^2 B_3 + 2B_2 B_3).$$

For completeness, the numerator identity needed here is

$$\begin{aligned} B_1^3 B_2 - 3B_1^2 B_3 + 2B_2 B_3 &= 2a(1+a^2)t^3 + 2(a^4 + 4a^2 + 3)t^4 - 2a(1+a^2)t^5 \\ &= 2t^3(1+a^2)\big[a + (a^2+3)t - at^2\big]. \end{aligned}$$

Substitution proves (7.3). □

Proof of Theorem 7.1. Take $\alpha = 4/3$ and $p = 1/12$. Then $a = -1$, $h = \pi/16$, and the third moment exists because $3p = 1/4 < 1$. We first verify the exact bound

$$t = \tan(\pi/16) < \frac{1}{5}.$$

Since $2 < 289/144$, we have $\sqrt{2} < 17/12$, and hence

$$\tan(\pi/8) = \sqrt{2} - 1 < \frac{5}{12} = \tan\bigl(2\arctan(1/5)\bigr).$$

Both arguments on the right and left lie in $(0, \pi/2)$. Strict monotonicity of tangent proves the asserted bound. It follows that

$$a + (a^2 + 3)t - at^2 = -1 + 4t + t^2 < -1 + \frac{4}{5} + \frac{1}{25} = -\frac{4}{25} < 0.$$

All other factors in (7.3) are positive. Thus $\kappa_3(Y) < 0$, contradicting the necessary condition in Lemma 7.2 for infinite divisibility. Self-decomposability implies infinite divisibility, so it fails as well.

The change of variables $x = y^{-12}$ in the density of $|C_{4/3}|$ gives the density in (7.2). The ratio of independent gamma variables with shapes $1/4$ and $3/4$ has beta-prime density $z^{-3/4}/(B(1/4, 3/4)(1 + z))$. Transforming by $y = z^{1/16}$, and using $B(1/4, 3/4) = \pi\sqrt{2}$, gives the same density. □

The factorization gives more than a single witness.

Theorem 7.4. *For every $1 < \alpha < 2$, define*

$$c = -\cot(\pi/\alpha) > 0, \qquad t_0(c) = \frac{2c}{c^2 + 3 + \sqrt{(c^2 + 1)(c^2 + 9)}}.$$

Then $|C_\alpha|^{-p}$ is not infinitely divisible whenever

$$0 < p < \frac{\alpha}{\pi} \arctan t_0(c). \tag{7.5}$$

Proof. With $a = -c$, the sign-controlling bracket in (7.3) is

$$ct^2 + (c^2 + 3)t - c.$$

Its positive root is $t_0(c)$, since $(c^2 + 3)^2 + 4c^2 = (c^2 + 1)(c^2 + 9)$; its other root is negative. It is therefore strictly negative for $0 < t < t_0(c)$. Moreover,

$$t_0(c) < \frac{c}{c^2 + 3} \le \frac{1}{2\sqrt{3}} < \tan(\pi/6).$$

Condition (7.5) implies $h < \pi/6 < \theta/3$, because $\theta = \pi/\alpha > \pi/2$. In particular $p < 1/3$, so Lemma 7.3 applies. Its third cumulant is strictly negative, and Lemma 7.2 completes the proof. □

Relation to later work and the scope of the conclusion. Wang [13, Section 1.3] repeats the negative-power question and proves sufficient conditions for infinite divisibility. For negative powers with $1 < \alpha \le 2$, the sufficient condition there is $p \ge \alpha/2$, which does not include the counterexamples above. Wang's later classification [14] concerns C_α itself: it is infinitely divisible exactly for $1 < \alpha \le 2$. Its introduction leaves the other Yano–Yano–Yor

questions for further research; that classification does not settle the negative powers considered here.

Bosch and Simon [12, final paragraph] also ask about infinite divisibility of the generalized beta-prime family $(G_b/G_a)^s$ for $s < 1$, after giving a sufficient subregion. The representation (7.2) places our example at $(a, b, s) = (3/4, 1/4, 1/16)$. This is a closely related application of the same witness, and is not counted as a second independent target.

Theorem 7.1 is a complete negative answer to the literal universal question in Remark 2.9(iii); it is not a classification of all (α, p). The proof uses the classical obstruction supplied by a negative cumulant. A bounded search of the original questions, related beta-prime power literature, and the cited subsequent papers found no identical resolution. This records the scope of the prior-art check; it does not establish exhaustive coverage or priority.

8 Verification and unresolved requirements

The stability argument is algebraic and applies to arbitrary finite elementary towers. The five moment and special-function arguments are analytic. The explicit Wright witness in Theorem 6.2 uses rational error bounds rather than a floating-point sign test. The negative third cumulant in Theorem 7.1 is certified by an exact trigonometric inequality. The total-positivity argument applies to arbitrary finite minors, and the monotonicity argument applies to every positive real argument. Finite experiments are not substituted for any of these quantifiers.

Separate internal AI derivations checked the principal algebraic identities, parameter domains, and convergence arguments. This is useful error checking, but it is neither external peer review nor formal proof-assistant verification. In particular, the manuscript should be read as a research draft whose arguments are exposed for scrutiny.

8.1 Literature coverage

For the Bell criterion, the original question was compared with Zhu's 2024 results [2], the exponential-Riordan criteria of Zhu [3], and the Stieltjes-moment results of Wang and Zhu [4]. Zhu's 2024 Proposition 6.18 establishes the pointwise Stieltjes property and the shifted-cumulant input, while its Conjecture 6.16(ii) still asks for coefficientwise positivity in the component variable. The distinction is essential. The proof here supplies its own positive measure for the input and does not depend on a displayed indexing formula in that article.

Dominici's author preprint for the Hankel-expansion paper [17] was also inspected. It treats discrete weights $\omega(k)z^k$ and leading Hankel determinants, with Vandermonde and Schur expansions; no generic shifted-cumulant criterion was found there. Differences between that preprint and every detail of the final article were not checked.

Two relevant full texts could not be inspected in this research session: Zhu's 2026 paper [16] and the general criterion in Ding–Mu–Zhu [18]. Their accessible metadata and previews are insufficient to exclude an equivalent theorem. Thus the general criterion and its application must not be announced as a verified world-first result on the basis of this draft.

For the Wright question, the original full text and subsequent work including Ferreira–Simon [15] were checked. For the negative-power question, the original remark, the related GB2 question of Bosch–Simon [12], and the stated ranges in Wang [13, 14] were compared. The full original Yang statement and its journal version were also checked. Kang's 28 October 2026 preprint [28] already refutes the separate Conjecture 1(ii), and its argument was inspected directly. That result is excluded from this draft's contribution count. Exact-title, conjecture, counterexample, and method searches located no identical resolution of the four older targets in the literature inspected. This records the extent of the search; it is not a proof that no earlier solution exists.

For elementary stability, the original Chen–Kauers question, Chen's 2022 stability paper, the Sahu–Seiss–Srinivasan generalized Liouville theorem, and the 2026 complete-reduction paper were compared. The 2026 paper of Sahu–Srinivasan [25] concerns elementary first integrals of triangular integral systems; no identical all-depth stability characterization was located there. Three independent internal reviews checked the local valuations, finite coefficient extension, logarithmic divisor argument, quotient-module descent, and finite-test classification. No fatal gap was found. This is a proposed complete proof for review, not an independently established novelty certificate.

For the all-order Yang problem, both the original preprint and final journal statement were checked, including the variable x and the prescribed decay. Orthogonal-polynomial Padé methods have substantial precedents: Prévost and Rivoal [29] give effective error estimates for nearly diagonal approximants to digamma asymptotic expansions. Only their abstract and first-page preview were accessible; the full paper remains a material priority-check gap. Li and Chen [30] treat Padé approximants for normalized gamma asymptotics; the complete-monotonicity lemma inspected there concerns ordinary Stirling truncations. Standard Stieltjes remainder identities give complete monotonicity in $s = x^2$, which does not imply the required property in x.

Section 5 supplies the additional matrix-marginal argument and its exact transform identity. The underlying Heine and Hermite–Genocchi identities, spectral Jacobian, Schur complement, determinant lemma, Bernstein representation, and gamma integral are classical ingredients. Two separate internal audits checked the full proof, including the gamma shape, rate normalization, bounded-transform interchanges, uniqueness, and every-order quantifier. A bounded search did not locate an identical all-order result or the exact intermediate matrix-marginal theorem; it does not certify their absence from the literature.

8.2 What remains open in this project

No unrestricted Erdős problem is claimed resolved here. The research also investigated graph inequalities, perfect-Skolem-set and permutation X-ray questions. Those investigations produced partial reductions or finite certificates, and they are not added to the six-target table. The formerly incomplete all-order DeTemple argument is now supplied in Section 5. The polynomial-weight dichotomy does not close Chen's full elementary-moment Skolem conjecture. Elementary repeated integration is treated separately by the proposed full characterization in Section 2. Similarly, instances, parameter families, equivalent formulations, and already known parts of a multipart conjecture are not counted as separate closures.

The requested ten-problem target, the preference for difficult longstanding problems, and the demand for a historically new method in every case remain unfulfilled. The differential-field descent, Bell-kernel criterion, and matrix marginal theorem are the principal proposed general results in this draft; no objective comparison with the value of the entire earlier corpus has been established.

The accompanying source is one self-contained `.tex` file with an embedded bibliography. It requires no external figures, data files, `.bib` file, or subsidiary `.tex` files. Run `pdflatex` until its internal-reference warnings disappear; the delivered version was compiled with three passes.

A A classical proof of Sokal's forest-triangle clause

This appendix concerns exactly Conjecture 6.11(a) of Sokal [1]. It does not resolve parts (b), (c), or the shifted Conjecture 6.12. The proof is an elementary composition of known totally positive matrices. Its inclusion records a complete mathematical argument, not a claim of a new foundational method or certified historical priority.

Let $T(z) = ze^{T(z)}$ be the tree function and define the exponential Riordan matrix

$$R[1, G]_{n,k} = \frac{n!}{k!}[z^n]G(z)^k.$$

Write $F = R[1, T]$ for the rooted-forest triangle, which is totally positive by Sokal's Theorem 1.1(a). The original conjecture asks whether

$$F^\sharp(w) = R\left[1, \frac{e^{wT(z)} - 1}{w}\right]$$

is coefficientwise totally positive in w. The expression at $w = 0$ is interpreted by its polynomial limit.

Proposition A.1. *The matrix $F^\sharp(w)$ is coefficientwise totally positive.*

Proof. Let $S(n, k)$ denote a Stirling number of the second kind and put $S_w(n, k) = S(n, k)w^{n-k}$ for $n \geq k$, and zero otherwise. The exponential generating function for $S(n, k)$ gives

$$S_w = R\left[1, \frac{e^{wz} - 1}{w}\right].$$

The ordinary Stirling matrix is totally positive. One standard verification uses $S(n, k) = h_{n-k}(1, 2, \ldots, k)$: these are path weights in a planar network with horizontal edges of weight j at level j and vertical edges of weight one. Sources $(-n, 1)$ and sinks $(-k, k)$ give the displayed complete homogeneous polynomial; the nonintersecting-path determinant identity gives nonnegative minors. The $k = 0$ row and column form a separate identity block.

For increasing index sets I, J of the same size, every nonzero term in the minor of S_w has the same power of w, so

$$\det(S_w)_{I,J} = w^{\sum_{i\in I} i - \sum_{j\in J} j} \det S_{I,J}.$$

If the exponent would be negative, every determinant term is zero by lower triangularity. Hence S_w is coefficientwise totally positive. Riordan multiplication now gives the exact factorization

$$F^\sharp(w) = FS_w.$$

Cauchy–Binet expresses every finite minor of this product as a finite sum of products of minors of F and S_w. Each summand has nonnegative coefficients, proving the claim. □

No explicit later resolution of this exact clause was located in the bounded literature check. Nevertheless, the method is classical, and Zhu [2] uses an analogous forest–Stirling factorization for functional digraphs. The two unread 2026 sources discussed above remain part of the priority check. This appendix must not be advertised as a high-difficulty new-method solution, or as a solution of all of Conjecture 6.11.

B Exact boundaries of the Bell-kernel method

The criterion in Section 3 concerns a fixed *real* cumulant sequence and coefficientwise positivity in one Bell parameter. The following exact examples rule out three tempting extensions. They were formulated while developing the present method; no pre-existing named open conjecture is identified with these extensions. Consequently none is counted toward the requested ten resolutions.

B.1 Polynomial cumulants do not give a coefficientwise extension

Let q be an indeterminate, and define $f_n(q)$ by the finite Stieltjes continued fraction

$$\sum_{n\geq 0} f_n(q)z^n = \cfrac{1}{1-\cfrac{z}{1-\cfrac{qz}{1-q^8z}}} = \frac{1-(q+q^8)z}{1-(1+q+q^8)z+q^8z^2}. \tag{B.1}$$

The nonnegative continued-fraction weights $1, q, q^8, 0, \ldots$ imply that $(f_n(q))$ is coefficientwise Hankel-totally positive. This follows from the standard weighted-Dyck-path and nonintersecting-path interpretation, so it is a formal polynomial assertion, not just positivity at fixed $q \geq 0$.

Set $\kappa_n(q) = f_{n-1}(q)$ for $n \geq 1$, and define

$$\sum_{n\geq 0} m_n(t,q)\frac{z^n}{n!} = \exp\left(t\sum_{n\geq 1}\kappa_n(q)\frac{z^n}{n!}\right).$$

Thus the shifted cumulants satisfy the polynomial analogue of the input condition in Section 3.

Proposition B.1. *The sequence $(m_n(t,q))$ is not coefficientwise Hankel-totally positive jointly in t, q.*

Proof. Equation (B.1) gives $f_0 = f_1 = 1$ and

$$f_n = (1+q+q^8)f_{n-1} - q^8 f_{n-2} \qquad (n \geq 2).$$

The exponential formula gives the exact Bell recurrence

$$m_0 = 1, \qquad m_n = t\sum_{j=1}^{n}\binom{n-1}{j-1}f_{j-1}m_{n-j}.$$

Applying these finite recurrences through $n = 6$ to the minor $D = m_4m_6 - m_5^2$ gives

$$\begin{aligned}[t^3]D = {} & 7q^{25} + 8q^{18} - 10q^{17} + 9q^{11} + 18q^{10} + 9q^9 \\ & + 8q^4 + 32q^3 + 48q^2 + 32q + 8.\end{aligned}$$

In particular, $[t^3q^{17}]D = -10$. This is a negative coefficient of a 2×2 Hankel minor, proving the assertion. □

For every *fixed* $q \geq 0$, the numerical hypothesis of Section 3 still applies. There is no contradiction: positivity of coefficients in t at each fixed q does not imply positivity of the individual coefficients in q.

B.2 Geometric mixing and factorial lifting can fail

Consider

$$K(z) = z + \frac{z^2}{1-21z} = \sum_{n\geq 1}\kappa_n\frac{z^n}{n!}, \qquad m_n(x) = n![z^n]\frac{1}{1-xK(z)}.$$

Here $\kappa_1 = 1$ and $\kappa_n = n!21^{n-2}$ for $n \geq 2$. Let G be the gamma probability measure of shape 2 and scale 21. The positive measure

$$\mu = \frac{20}{21}\delta_0 + \frac{1}{21}G$$

represents $(\kappa_{n+1})_{n\geq 0}$: for $n \geq 1$, $\int u^n\,\mathrm{d}\mu(u) = (n+1)!21^{n-1} = \kappa_{n+1}$, and the zeroth moment is one.

Proposition B.2. *The shifted cumulants in this example form a numerical Stieltjes moment sequence, and $(m_n(x)/n!)$ is coefficientwise Hankel-totally positive, but $(m_n(x))$ is not coefficientwise Hankel-totally positive.*

Proof. The moment representation was just verified. The normalized generating function has the positive Stieltjes fraction

$$\frac{1}{1-xK(z)} = \cfrac{1}{1-\cfrac{xz}{1-\cfrac{z}{1-20z}}},$$

which proves the coefficientwise assertion for $(m_n/n!)$. For the unnormalized sequence, direct coefficient extraction gives

$$\begin{aligned} &m_1 = x, && m_2 = 2x^2 + 2x, \\ &m_3 = 6x^3 + 12x^2 + 126x, \\ &m_4 = 24x^4 + 72x^3 + 1032x^2 + 10584x, \\ &m_5 = 120x^5 + 480x^4 + 7920x^3 + 110880x^2 + 1111320x. \end{aligned}$$

Expanding the shifted leading minor yields

$$\begin{aligned} \det(m_{i+j+1})_{i,j=0}^{2} = 24x^3\big(&x^6 + 6x^5 + 195x^4 - 860x^3 \\ &+ 86295x^2 + 79926x + 1120581\big). \end{aligned}$$

The coefficient of x^6 is $-20640 < 0$. □

The same construction with $K_b(z) = z + z^2/(1-bz)$, $b \geq 1$, gives

$$[x^6] \det(m_{i+j+1})_{i,j=0}^{2} = 96(-b^2 + 11b - 5).$$

Thus the failure is systematic. It does not refute Sokal's Conjecture 6.7, which fixes the specific tree function $K = T$. The already known failure of general coefficientwise Hadamard closure also cautions against factorial lifting; the example here supplies an explicit certificate inside the proposed geometric construction.

B.3 A nested exponential formula also needs extra hypotheses

Keep the same numerical K, and define

$$H_w(z) = \frac{e^{wK(z)} - 1}{w}, \qquad \sum_{n \geq 0} M_n(t,w)\frac{z^n}{n!} = e^{tH_w(z)}.$$

Although the innermost shifted cumulants are numerical Stieltjes moments, the nested formula need not give joint coefficientwise Hankel positivity. Indeed, with $D = \det(M_{i+j+1}(t,w))_{i,j=0}^{2}$, one obtains

$$[t^4]D = 2w^4 - 82w^3 - 5144w^2 + 1035540w + 959112.$$

For completeness, this identity can be checked without any infinite series manipulation. Start with

$$(\kappa_1, \ldots, \kappa_5) = (1, 2, 126, 10584, 1111320),$$

form $b_0 = 1$ and $b_n = w\sum_{j=1}^{n}\binom{n-1}{j-1}\kappa_j b_{n-j}$, put $h_n = b_n/w$, and then form $M_0 = 1$ and $M_n = t\sum_{j=1}^{n}\binom{n-1}{j-1}h_j M_{n-j}$. These recurrences through $n = 5$ give the displayed determinant coefficient. The negative t^4w^2 coefficient is -5144. This rules out a general nested-Bell shortcut, but does not decide Sokal's forest conjectures, which have additional structure.

B.4 Status of the continuation

The latest continuation produced the all-order complete-proof candidate for Yang–Chu–Zhang Problem 1 in Section 5, and strengthened the elementary-stability argument with the polynomial-weight dichotomy. These are one additional original target and one general structural strengthening, respectively. The draft has not established ten new resolutions or certified all historical-priority and new-method requirements. Appendix A provides a complete but classical argument for one exact published clause. The three obstructions above are proved statements about attempted extensions of our method. Neither of those ancillary categories is relabeled as a new difficult named-conjecture resolution. The multivariate Hankel claims, Chen's full moment-set conjecture, and the unrestricted Erdős targets remain unresolved in this work.

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